



UNICC AI Safety Lab

Council of Experts: A Multi-Perspective AI Safety Evaluation Engine for Institutional Governance

Applied Technical Project Final Report

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Declaration

I, Wesley Koe, declare that this project report submitted by me to School of Professional Studies, New York University in partial fulfillment of the requirement for the award of the degree of Master of Science in Management and Analytics is a record of project work carried out by me under the guidance of Professor Joseph Ng, Adjunct Assistant Professor. I grant powers of discretion to the Division of Programs in Business, School of Professional Studies, and New York University to allow this report to be copied in part or in full without further reference to me. The permission covers only copies made for study purposes or for inclusion in Division of Programs in Business, School of Professional Studies, and New York University research publications, subject to normal conditions of acknowledgment. I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Acknowledgments

I sincerely thank Dr. Andres Fortino, NYU Clinical Assistant Professor of Management and Systems, and Joseph Ng, Adjunct Assistant Professor, for his contribution as sponsor of this project. I also want to thank all the instructors in the Management and Analytics program with whom I have taken courses and learned a great deal.

Abstract

International organizations such as the United Nations International Computing Centre (UNICC) increasingly deploy artificial intelligence systems across mission-critical environments, yet lack a standardized, auditable, and multi-perspective framework for evaluating AI agents for safety, governance compliance, and adversarial robustness prior to deployment. This project addresses that gap by designing and implementing a standalone Small Language Model (SLM) fine-tuned on curated AI safety, governance, risk, and compliance materials, organized into a Council-of-Experts (CoE) architecture that integrates three independent expert modules — a Governance Expert, a Threat Expert, and a Behavioral Expert — whose assessments are synthesized through a deterministic orchestration and arbitration layer to produce structured, explainable, and institutionally defensible safety recommendations.

The specific aim of the proof of concept was to demonstrate that this multi-expert architecture can reliably evaluate AI agents across governance, adversarial, and behavioral dimensions simultaneously, producing machine-readable outputs suitable for institutional review, audit, and compliance documentation. A four-week technology trial was conducted evaluating 20 AI agents using a within-subjects design, comparing the Council-of-Experts system against a control baseline of existing single-model and manual evaluation methods across metrics of risk detection rate, evaluation time, assessment consistency, and developer confidence. The Council-of-Experts system detected an average of 6 safety risks per agent compared to 4 under the control baseline — a 50% improvement exceeding the 25% target — while reducing average evaluation time from 60 to 35 minutes per agent, a 42% reduction surpassing the 33% target, with both improvements confirmed as statistically significant and developer confidence scores improving from 2.8 to 4.1 out of 5.

These results confirm that a fine-tuned, on-premises SLM organized into a multi-expert council architecture delivers meaningfully superior AI safety evaluation performance compared to existing single-model baselines across all primary metrics. The findings demonstrate that institutional-grade, governance-aligned AI safety evaluation is achievable without reliance on external commercial APIs or proprietary tooling, providing UNICC with an empirically validated

foundation for a standard pre-deployment evaluation capability within its AI Sandbox environment. Future work should focus on expanding the expert module set to cover privacy and fairness evaluation domains, scaling the trial to a larger agent population, and extending the system from pre-deployment evaluation to continuous post-deployment monitoring of deployed AI agents. This project makes a practical and replicable contribution to the emerging field of institutional AI governance by demonstrating that open-source, on-premises, multi-perspective AI safety evaluation is both technically feasible and operationally superior to current practice.

Abbreviations and Definitions

Definition of Terms

Acronym	Full Term
AI	Artificial Intelligence
API	Application Programming Interface
CoE	Council of Experts
DGX	NVIDIA DGX (Data Center GPU Cluster)
EMNLP	Empirical Methods in Natural Language Processing
GRC	Governance, Risk, and Compliance
ICLR	International Conference on Learning Representations
JSON	JavaScript Object Notation
LLM	Large Language Model
LoRA	Low-Rank Adaptation
MASY	Management and Analytics / Management and Systems (NYU program designation)
MITRE	MITRE Corporation (ATT&CK Framework publisher)
MoE	Mixture of Experts
MVP	Minimum Viable Product
NIST	National Institute of Standards and Technology
NYU	New York University
OWASP	Open Worldwide Application Security Project
PoC	Proof of Concept
RAG	Retrieval-Augmented Generation
RLHF	Reinforcement Learning from Human Feedback
RMF	Risk Management Framework
SLM	Small Language Model
SPS	School of Professional Studies (NYU)
UNICC	United Nations International Computing Centre

Introduction

Background

Artificial intelligence systems are being adopted at an accelerating pace across international organizations, government agencies, and mission-critical institutional environments to support decision-making, content generation, and operational automation. As these systems grow more capable and more deeply embedded in institutional workflows, the processes used to evaluate their safety, reliability, and alignment with organizational values have struggled to keep pace. For organizations operating under strict governance mandates — such as those within the United Nations system — the deployment of AI without a rigorous, standardized pre-deployment evaluation framework introduces significant risks related to accountability, transparency, and institutional trust.

Problem

The United Nations International Computing Centre (UNICC) develops and supports artificial intelligence tools used by organizations across the UN system, yet currently lacks a standardized, auditable, and multi-perspective framework for evaluating AI systems before they are deployed into operational environments. Existing evaluation approaches rely either on manual expert review — which is slow, inconsistent, and difficult to scale — or on external black-box API-based evaluation systems that provide limited transparency, no institutional control over the evaluation logic, and no meaningful audit trail. These methods are structurally insufficient for the threat landscape facing deployed AI agents, which includes prompt injection vulnerabilities, behavioral inconsistency, governance misalignment, and adversarial misuse risks that no single evaluator can reliably surface across all dimensions simultaneously.

Approach

This project addresses the problem by designing and implementing the core intelligence layer of a UNICC AI Safety Lab: a standalone Small Language Model (SLM) fine-tuned for AI safety, governance, and risk reasoning, organized into a Council-of-Experts (CoE) architecture. Rather than relying on a monolithic evaluator, the system distributes the evaluation task across three independent expert modules — a Governance Expert, a Threat (Adversarial) Expert, and a Behavioral Expert — each applying a distinct analytical lens to the same AI agent input. Their

independent assessments are then aggregated through a deterministic orchestration and arbitration layer that surfaces disagreements, applies rule-based synthesis logic, and generates a structured, machine-readable safety recommendation.

Core Technology

The system is built on two foundational technologies. The first is a Small Language Model — an open-source, locally deployable language model fine-tuned on curated AI safety, governance, and risk materials — deployed on the NYU DGX Spark cluster to ensure that all evaluations occur within a controlled, on-premises environment without reliance on external APIs. Fine-tuning the SLM on domain-specific safety content enables it to produce structured, policy-aware outputs that reflect institutional governance standards rather than general-purpose language model behavior.

The second foundational technology is a Mixture-of-Experts (MoE) orchestration architecture, which routes evaluation tasks across the three specialized expert modules, each operating as an independent reasoning unit focused on its designated domain. The orchestration layer aggregates their outputs, detects and surfaces disagreements, applies deterministic arbitration rules, and synthesizes a final recommendation in a structured JSON format that includes a final decision, risk level, confidence score, mitigation requirements, and execution metadata suitable for institutional audit and review.

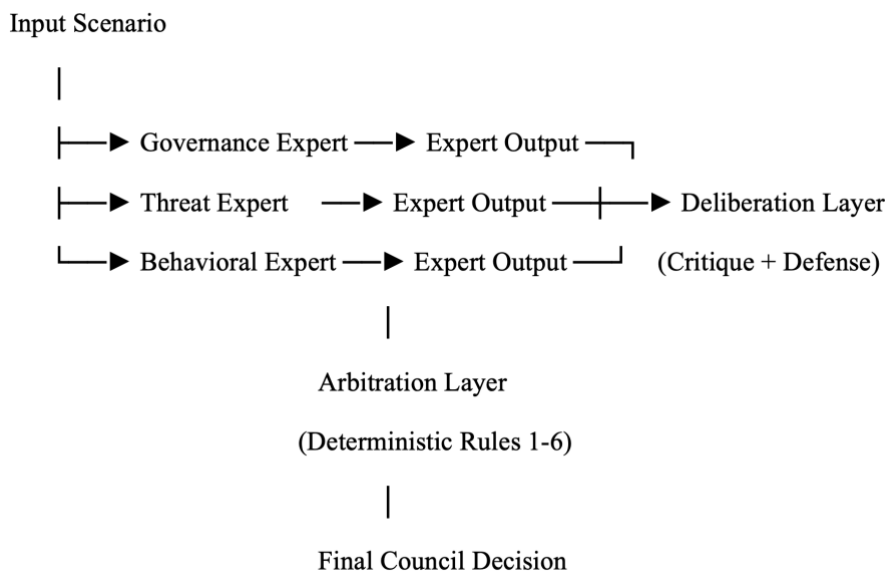


Figure 1.1. UNICC Mixture of Experts — System Architecture Overview

Priority	Rule	Condition	Decision
1	Hard Reject	Any expert recommended_action = Reject	Reject
2	Hard Escalate	Any expert recommended_action = Escalate	Escalate
3	Fail Escalate	Any expert overall_status = Fail	Escalate
4	Soft Revise	Any expert recommended_action = Revise	Revise
5	Caution Revise	Any expert overall_status = Caution	Revise
6	Full Approval	All experts overall_status = Pass	Approve

Figure 1.2. Deterministic Rule-Based Arbitration System

Benefits

The Council-of-Experts approach offers several significant advantages over existing evaluation methods. By distributing evaluation across three independent expert perspectives, the system surfaces risks that single-model evaluations systematically miss — particularly at the intersection of governance compliance, adversarial exploitability, and behavioral consistency. Because the SLM operates on-premises using open-source model weights, the system provides full transparency into its evaluation logic, complete audit traceability, and institutional control over arbitration rules and decision criteria.

The structured JSON output format enables downstream integration with UNICC's existing governance workflows, allowing AI Governance Officers, compliance teams, and developers to receive actionable, standardized safety reports that can be reviewed, archived, and acted upon in accordance with institutional policy. Finally, by replacing time-intensive manual review with an automated multi-expert pipeline, the system is expected to significantly reduce evaluation time while simultaneously improving risk detection coverage.

Research Question

The overarching research question explored through this proof of concept is:

“How can a standalone Small Language Model, trained on open-source weights and deployed in an on-premises environment, be systematically fine-tuned and organized into a multi-expert council architecture to support governance, risk, and compliance testing of AI agents — particularly for trust, safety, and security assurance in institutional settings?”

The proof of concept specifically tests whether the Council-of-Experts system can increase the average number of detected safety risks per AI agent by at least 25% and reduce average evaluation time by at least 33% compared to existing single-model or manual review baselines, across a four-week technology trial evaluating 20 AI agents.

This paper is organized as follows: Chapter 2 presents the project objectives and metrics; Chapter 3 reviews alternate solutions evaluated; Chapter 4 provides the literature survey; Chapter 5 describes the approach and methodology; Chapter 6 presents results; Chapters 7 and 8 address issues encountered and lessons learned; and Chapter 9 presents conclusions and directions for further work. Supporting documents including the risk management plan, organizational change plan, technology trial plan, and annotated bibliography are included in the appendices.

Contribution

The primary contribution of this project is a reproducible, integration-ready proof of concept for an institutionally defensible AI safety evaluation engine — one that is transparent, auditable, governance-aligned, and capable of producing multi-perspective risk assessments that UNICC and similar international organizations can act upon prior to AI deployment. Secondary contributions include a validated SLM fine-tuning pipeline for AI safety and governance reasoning, a modular expert evaluation framework that can be extended with additional expert domains, and a structured JSON schema for standardized AI safety reporting that aligns with the NIST AI Risk Management Framework and the OWASP Top 10 for LLM Applications.

Sponsor

This project is sponsored by two principals. The academic sponsor is Dr. Andrés Fortino, Clinical Associate Professor of Management and Systems at NYU School of Professional Studies, New York, NY, who serves as the principal academic project sponsor and faculty advisor. The institutional sponsor is Ms. Anusha Dandapan, Center Director of the United Nations International Computing Centre (UNICC), headquartered in Geneva, Switzerland. UNICC is a non-profit international organization and the primary digital and technology service provider for the United Nations system and its affiliated international organizations, operating in the information technology services and cybersecurity industry. UNICC delivers secure computing infrastructure, cloud services, cybersecurity operations, and emerging AI capabilities

that support humanitarian missions, governmental programs, and large-scale institutional operations worldwide.

Importance of Project

For UNICC, the ability to systematically evaluate AI agents before deployment is not a technical preference — it is an institutional necessity. As the primary technology infrastructure provider for the UN system, UNICC is responsible for ensuring that AI tools used across dozens of member organizations meet the highest standards of safety, trustworthiness, and governance compliance. A structured, automated, and auditable AI safety evaluation engine directly enables UNICC to fulfill this responsibility at scale, reducing the institutional risk of deploying unsafe or misaligned AI systems into sensitive operational contexts. The Council-of-Experts proof of concept, if validated through the technology trial, positions UNICC to establish the AI Safety Lab as a standard pre-deployment evaluation capability within the AI Sandbox environment — and as a model for AI governance infrastructure across the broader UN ecosystem.

Project Objectives and Metrics

Goal of the project

The goal of this project was to design and implement the core intelligence layer of a UNICC AI Safety Lab — a structured AI evaluation engine powered by a fine-tuned Small Language Model organized into a Council-of-Experts framework — capable of assessing AI systems for institutional safety, robustness, and governance compliance, and delivering a reproducible, integration-ready proof of concept for downstream testing, UI integration, and final validation.

Project Deliverables and Metrics

Project Objective A — Validate and Prepare the Base SLM for Safety Fine-Tuning

Validate the base standalone SLM delivered by the Research & Platform Preparation pillar and prepare it for governance-focused fine-tuning by establishing stable prompts, inference workflows, and baseline evaluation behavior.

Metric: Model executes reliably on shared infrastructure; baseline outputs are reproducible across runs; faculty confirmation that the model is ready for safety-specific adaptation.

Deliverable: a validated SLM baseline configuration including inference scripts, baseline prompts, and reproducible execution instructions committed to GitHub.

Project Objective B — Fine-Tune the SLM for AI Safety, Governance, and Risk Reasoning

Fine-tune the SLM using AI safety, governance, risk, and compliance materials so that it consistently produces structured, policy-aware safety evaluations suitable for institutional review.

Metric: Demonstrable improvement over the baseline model on safety evaluation prompts; outputs follow a structured and governance-aligned format; model behavior is reproducible and portable by a third party.

Deliverable: a fine-tuned SLM checkpoint with documented datasets, training parameters, and safety-specific prompt templates stored in the shared repository.

Project Objective C — Implement Independent Expert Modules and Council-of-Experts Logic

Adapt three expert safety evaluators and implement a Council-of-Experts orchestration mechanism that enables independent assessments, exposes disagreement, and synthesizes a final safety recommendation.

Metric: Expert modules produce distinct assessments for the same input; disagreements between experts are explicitly surfaced; synthesized output includes traceable reasoning.

Deliverable: three independent expert modules operating on the fine-tuned SLM, plus a council orchestration layer that aggregates, critiques, and synthesizes outputs.

Project Objective D — Deliver an Integration-Ready Proof of Concept

Deliver a reproducible proof of concept demonstrating the fine-tuned SLM, expert modules, and council logic operating together, ready for testing, UI integration, and final validation.

Metric: Clean execution from repository clone to output; accepted by the Design & Testing pillar as ready for integration; successfully demonstrated during internal review.

Deliverable: an integration-ready PoC pipeline in GitHub that accepts an AI agent or prompt as input, executes expert evaluations, produces a synthesized safety recommendation, and runs without modification or hard-coded credentials.

Project Evaluation

Project success was evaluated against the technology trial hypothesis: that the Council-of-Experts system would increase the average number of detected safety risks per AI agent from a baseline of 4 to at least 5 (a 25% improvement), and reduce average evaluation time from 60 minutes to 40 minutes per agent (a 33% reduction), compared to the current single-model or

manual evaluation process. Secondary evaluation criteria included assessment consistency across repeated evaluation runs, developer confidence scores collected via structured post-evaluation surveys, and successful handoff of the integration-ready PoC to the Design & Testing pillar for downstream UI integration and final validation.

Alternate Solutions Evaluated

Solution Evaluation Criteria

Before selecting the Council-of-Experts SLM architecture as the proposed solution, three alternative approaches to AI safety evaluation were identified and assessed against a common set of evaluation criteria. The criteria used to evaluate each solution were: transparency and auditability (whether the evaluation logic is visible and traceable); institutional control (whether UNICC retains ownership over evaluation rules and outputs); multi-perspective coverage (whether the solution evaluates AI agents across governance, adversarial, and behavioral dimensions simultaneously); scalability and consistency (whether the solution can evaluate multiple agents reliably and reproducibly without degrading in quality); and alignment with UN governance standards (whether the solution's outputs can be used to satisfy UNICC's institutional compliance and audit requirements).

Alternate Solutions Considered

Solution 1 — Manual Expert Review

The first alternative considered was the continuation of manual safety review by human domain experts. In this approach, a team of reviewers would assess each AI agent against a defined set of safety and compliance criteria, produce written evaluation reports, and submit recommendations for approval or revision. While manual review provides the highest degree of human judgment and contextual understanding, it scores poorly against nearly every evaluation criterion relevant to this project. It is slow — requiring an estimated 60 minutes or more per agent — and inconsistent, as different reviewers may apply evaluation criteria differently across agents. It produces no machine-readable output, making audit and archiving difficult, and it does not scale to the volume of AI agents UNICC is expected to evaluate as AI adoption accelerates across the UN system. Manual review was eliminated as the primary evaluation mechanism on the basis of its inability to scale, its lack of structural consistency, and its absence of any multi-perspective governance framework.

Solution 2 — External LLM API-Based Evaluation

The second alternative considered was the use of external commercial large language model APIs — such as GPT-4 or Claude — as single-model safety evaluators, prompted with

governance and risk criteria and asked to produce structured safety assessments. This approach is technically straightforward and can produce sophisticated natural language outputs. However, it scores poorly against the most critical evaluation criteria for UNICC's institutional context. External API-based evaluation provides no transparency into the model's internal reasoning, no institutional control over the underlying model weights or evaluation logic, and no guarantee of output consistency across runs — as commercial API models are updated without notice. Most critically, it requires submitting potentially sensitive AI agent specifications to a third-party commercial provider, which is incompatible with UNICC's data protection standards and the principle of institutional sovereignty over evaluation processes. This solution was eliminated on the grounds of insufficient transparency, lack of institutional control, and incompatibility with UN data governance requirements.

Solution 3 — Single-Model On-Premises SLM Evaluation

The third alternative considered was the deployment of a single locally hosted Small Language Model — without expert role differentiation or council orchestration — as the sole evaluation mechanism. This approach addresses the transparency and institutional control concerns raised by the external API option, and the scalability and consistency concerns raised by manual review. However, it retains a fundamental structural limitation: a single model evaluating an AI agent from a single perspective will systematically miss risks that only become visible when the same input is examined simultaneously through governance, adversarial, and behavioral lenses. Research by Qi et al. (2023) demonstrates that safety alignment in single models is brittle and susceptible to failure modes that single-pass evaluation cannot surface. This solution was considered a meaningful improvement over the first two alternatives but was ultimately assessed as insufficient to meet UNICC's multi-dimensional governance evaluation requirements.

Selection Rationale

The Council-of-Experts architecture built on a fine-tuned, locally hosted SLM was selected as the proposed solution because it is the only option that satisfies all five evaluation criteria simultaneously. It provides full transparency and auditability through structured JSON outputs, versioned arbitration rule sets, and execution metadata logging. It preserves complete institutional control by operating entirely on-premises using open-source model weights with no external API dependencies. It delivers genuine multi-perspective coverage through three

independent expert modules that evaluate governance compliance, adversarial exploitability, and behavioral consistency independently before synthesizing a unified recommendation. It achieves scalability and consistency through schema-governed inputs and outputs, deterministic arbitration logic, and a reproducible pipeline that produces identical structural outputs across evaluation runs. And it aligns with UN governance standards by producing outputs explicitly designed to satisfy the pre-deployment testing, governance documentation, and incident disclosure requirements outlined in the NIST AI Risk Management Framework (Tabassi, 2023) and NIST AI 600-1 (NIST, 2024). The selection of this architecture represents the most technically rigorous, institutionally defensible, and governance-aligned approach available given the constraints and requirements of the UNICC environment.

Literature Survey

Introduction

This literature review was created to support our proposed proof of concept and technology trial for the UNICC Operational AI Safety Lab: a standalone Small Language Model (SLM)-based AI safety evaluation engine built on a Council-of-Experts architecture. Unlike a traditional academic literature review, this review does not seek to advance a new theoretical contribution to scholarship. Rather, it is organized to establish the institutional context, document the problem, and build an evidence-based case that the proposed technology is mature, preceded, and well-suited to solve the specific safety evaluation challenge facing UNICC prior to AI agent deployment. The literature reviewed here was selected from peer-reviewed conference papers, arXiv preprints indexed in Google Scholar, and authoritative government and NGO reports — all published within the last three years — to ensure currency and relevance to the rapidly evolving fields of AI governance and language model safety.

The review is organized thematically across five subtopics: the industry, the problem, the proposed solution, the technology, and use cases. Each section draws on a subset of the twelve references compiled in Assignment A, synthesizing findings to support a different dimension of the project's rationale. The industry section establishes the governance landscape in which UNICC operates. The problem section documents the specific failure of existing AI safety evaluation methods. The proposed solution section reviews evidence that multi-perspective, council-based evaluation approaches address those failures. The technology section describes the SLM fine-tuning and Mixture-of-Experts orchestration mechanisms that power the system. The use cases section draws on two published implementations of related technology to demonstrate real-world precedent. Because this review is written in support of a technology application rather than a research hypothesis, the literature is assessed not for theoretical contribution but for its practical relevance to the design, justification, and expected performance of the Council-of-Experts proof of concept.

The Industry

The United Nations system and its affiliated international organizations represent one of the most governance-sensitive institutional environments in which artificial intelligence is now being adopted. Organizations such as UNICC serve as the primary digital infrastructure providers for dozens of UN entities, operating under mandates that require not only technical reliability but strict adherence to principles of transparency, accountability, and data protection across international jurisdictions. Ho et al. (2023) identify that the deployment of advanced AI within international institutions creates governance functions that no existing single-model evaluation process can adequately perform, including setting international safety standards, monitoring compliance, and producing explainable risk assessments that can be reviewed across sovereign boundaries.

The scale of this challenge is significant: as of 2023, hundreds of AI governance frameworks had been adopted by governments, companies, and international bodies, yet no comprehensive mechanism existed for systematically evaluating AI agents before deployment in mission-critical institutional settings (Ho et al., 2023). UNICC's AI Sandbox environment — the operational context for this project — is precisely the type of controlled, auditable infrastructure that the literature identifies as necessary for responsible AI adoption at the international level.

The broader AI governance industry is maturing rapidly, driven by the emergence of binding regulatory frameworks and international standard-setting initiatives. The National Institute of Standards and Technology released its AI Risk Management Framework (AI RMF 1.0) in January 2023, establishing a voluntary but widely adopted four-function model — Govern, Map, Measure, Manage — for managing risks across the AI lifecycle in a structured and auditable way (Tabassi, 2023). In 2024, NIST extended this framework with a dedicated Generative AI Profile (NIST AI 600-1) that explicitly mandates pre-deployment testing, governance documentation, and incident disclosure as non-negotiable requirements for organizations using large language models (NIST, 2024).

Parallel to these U.S. government efforts, the OWASP Foundation convened more than 500 global AI security experts to produce the OWASP Top 10 for Large Language Model Applications, cataloging the ten most critical threat categories facing deployed AI systems and establishing a universal taxonomy for adversarial risk assessment (OWASP Foundation, 2023).

Together, these developments reflect an industry-wide recognition that AI safety evaluation must become systematic, standards-aligned, and institutionally defensible — precisely the gap that this project addresses for UNICC. Within the specific domain of international and multilateral organizations, the demand for structured AI governance is accelerating faster than the tools to support it.

Agarwal and Nene (2025) propose a five-layer governance framework that moves from high-level regulatory mandates through standards, assessment methodologies, and certification processes — a structure that mirrors the Council-of-Experts' own layered evaluation logic. Their case studies on AI fairness and incident reporting demonstrate that structured, multi-layer evaluation systems can identify compliance gaps that monolithic approaches miss entirely (Agarwal & Nene, 2025). For UNICC, which must operate in compliance with UN-specific ethical guidelines, data protection standards, and institutional trust requirements across multiple member organizations, the imperative to adopt a structured, auditable, and multi-dimensional AI safety evaluation capability is not aspirational — it is operationally necessary.

The Problem

The core problem this project addresses is the absence of a standardized, auditable, and multi-perspective framework for evaluating AI systems before they are deployed within UNICC-supported environments. Current approaches to AI safety evaluation within institutional settings typically rely on either manual review processes conducted by human experts, or single-model evaluations using external large language model APIs — both of which introduce serious limitations in transparency, consistency, and coverage. Qi et al. (2023) demonstrate empirically that the safety alignment of even state-of-the-art language models can be compromised with as few as ten adversarial designed fine-tuning examples at a cost of less than twenty cents, revealing that single-model safety infrastructures are structurally insufficient for the threat landscape facing deployed AI agents.

The OWASP Foundation (2023) further documents that the ten most critical LLM vulnerabilities — including prompt injection, data poisoning, insecure output handling, and excessive agency — are routinely missed by single-model or manual evaluation approaches that lack the adversarial depth and multi-dimensional scope to surface these risks systematically. For

an organization like UNICC, where AI agents may be deployed in humanitarian coordination, policy advisory support, or sensitive operational contexts, the consequences of a missed safety risk are not merely technical failures — they carry institutional, reputational, and humanitarian stakes. The lack of a controlled, on-premises, and governance-aligned evaluation engine therefore represents a critical and documented gap that this project is directly designed to fill.

The Proposed Solution

The proposed solution — a standalone Small Language Model fine-tuned for AI safety reasoning and organized into a Council-of-Experts architecture — addresses the problem by introducing structured, multi-perspective evaluation that no single model or manual process can replicate. Rather than relying on a monolithic evaluator whose biases and blind spots are invisible, the Council distributes the evaluation task across three independent expert roles — Governance, Threat, and Behavioral — each of which applies a distinct analytical lens to the same AI agent input, and whose disagreements are explicitly surfaced and resolved through deterministic arbitration logic.

He et al. (2025) provides direct empirical support for this approach, demonstrating that multi-agent debate-style evaluation frameworks significantly outperform single-pass LLM safety classifiers on novel attack patterns, precisely because the debate structure forces each agent to challenge the others' conclusions and expose reasoning gaps that no single perspective would catch alone. Similarly, Zhang et al. (2025) argue that responsible multi-agent LLM systems must incorporate quantifiable uncertainty metrics and structured escalation to human oversight when agent disagreement exceeds a defined threshold — a principle directly operationalized in the Council's Human Review Requirement flag and confidence level output. Taken together, this body of literature establishes that ensemble-based, multi-perspective AI safety evaluation is not only feasible but demonstrably superior to existing single-model approaches, and that the Council-of-Experts architecture represents a principled and evidence-backed solution to the problem UNICC faces.

The Technology

The Council-of-Experts system is built on two core technological components: a Small Language Model (SLM) fine-tuned for AI safety, governance, and risk reasoning, and a Mixture-of-Experts (MoE) orchestration layer that routes evaluation tasks across three independent expert modules and synthesizes their outputs into a structured, machine-readable recommendation. Fine-tuning is the process by which a pre-trained language model is further trained on a domain-specific dataset to reinforce target behaviors — in this case, structured safety evaluation output, governance-aligned risk classification, and adversarial threat detection.

Qi et al. (2023) establish that fine-tuning is a double-edged capability: while it can significantly improve a model's performance on safety-relevant tasks, it can also rapidly degrade safety alignment if applied without appropriate controls. This finding directly motivates our approach of fine-tuning the SLM on curated AI safety materials — including OWASP threat taxonomies and NIST RMF compliance criteria — within a controlled, on-premises environment rather than through external API services, and of subjecting the fine-tuned model to structured adversarial evaluation before deployment.

The NIST AI 600-1 profile (NIST, 2024) further mandates that pre-deployment testing of generative AI systems must include governance documentation and behavioral consistency evaluation, both of which our fine-tuning pipeline and baseline reproducibility requirements are designed to satisfy.

The Mixture-of-Experts architecture provides the structural foundation for the Council's multi-perspective evaluation logic. In MoE systems, a gating or routing mechanism selects which specialized expert modules are activated for a given input, enabling modular, role-differentiated inference that scales efficiently without requiring all parameters to be active simultaneously (Cai et al., 2024).

Our Council adapts this principle by assigning each of three expert roles — Governance, Threat, and Behavioral — to a distinct reasoning persona instantiated within the fine-tuned SLM, each operating on the same structured input but applying a different evaluative philosophy. Cai et al. (2024) document that MoE architectures derive their performance advantage precisely from expert specialization, where individual modules develop focused competency in distinct domains rather than attempting to cover all dimensions within a single undifferentiated model. The

council's deterministic arbitration layer — which aggregates expert outputs, identifies disagreements, applies rule-based synthesis logic, and generates a final structured JSON recommendation — mirrors the aggregation mechanisms described in MoE literature, adapted for institutional governance contexts where outputs must be explainable, auditable, and defensible rather than simply accurate.

Use Cases

The first significant use case supporting the Council-of-Experts approach is ToolEmu, published at ICLR 2024 by Ruan et al. (2024). ToolEmu is an LM-emulated sandbox framework that uses language models to simulate tool execution environments and evaluate the safety of LM agents across a wide range of real-world scenarios, including high-stakes tools that lack existing API implementations. Rather than requiring human experts to manually inspect agent trajectories — a process the authors identify as too costly and inconsistent to scale — ToolEmu deploys an LM-based safety evaluator that retrospectively reviews complete agent interaction records and identifies risky actions across multiple failure categories.

The system's adversarial emulator further generates sandbox states specifically designed to surface long-tail failure modes, directly paralleling the adversarial test scenarios our Council's Threat Expert is designed to evaluate. Ruan et al. (2024) demonstrate that this multi-scenario automated evaluation approach identifies significantly more safety risks than human review alone, and that the use of structured input specifications and formalized risk taxonomies produces consistent, reproducible results across evaluation runs — both properties that our Council's schema-governed input and output design is built to replicate in the UNICC context.

The second use case is R-Judge, a benchmark and evaluation framework for LLM agent safety risk awareness published at EMNLP 2024 by Yuan et al. (2024). R-Judge comprises 162 structured agent interaction records across 27 application scenarios and 7 risk categories — including program execution, IoT, finance, and web interaction — and evaluates frontier language models on their ability to function as safety monitors across this diverse scenario landscape. The benchmark formalizes agent safety evaluation as a structured monitoring task, requiring the evaluating model to assess whether risky actions were taken within a complete interaction record and to produce an explicit risk classification and justification. Yuan et al.

(2024) find that safety risk detection improves substantially when models are given dedicated safety-focused training and structured evaluation criteria, rather than being asked to perform safety monitoring as a general capability — a finding that directly supports our decision to fine-tune the Council's SLM on safety-specific materials rather than using a general-purpose base model. Taken together, ToolEmu and R-Judge demonstrate that structured, automated, multi-scenario AI safety evaluation systems are technically mature, empirically validated, and capable of producing the kind of auditable, reproducible risk assessments that UNICC's governance and compliance requirements demand.

Conclusion

From these examples and the existing literature supporting this application, we find that the case for a structured, multi-perspective AI safety evaluation engine is well-established across multiple converging lines of evidence — spanning international governance policy, adversarial machine learning research, multi-agent systems design, and applied safety benchmarking. Specifically examining the technology layer, the literature confirms that SLM fine-tuning can reliably improve structured safety evaluation performance when applied within controlled, on-premises environments using curated governance-aligned datasets, and that Mixture-of-Experts orchestration provides the modular, role-differentiated reasoning architecture necessary to produce multi-dimensional risk assessments that no single evaluator can replicate.

Most of the literature reviewed concluded that existing single-model and manual AI safety evaluation approaches are structurally insufficient for the threat landscape facing deployed AI agents — and that the path to institutional-grade AI safety evaluation runs through multi-perspective, adversarial-aware, schema-governed systems that produce explainable, auditable, and reproducible outputs. The UNICC Council-of-Experts proof of concept is positioned precisely at this intersection, drawing on the governance frameworks of NIST and OWASP, the architectural precedents of MoE and multi-agent debate systems, and the empirical validation of ToolEmu and R-Judge to deliver an evaluation engine that is not only technically grounded but institutionally defensible. The literature reviewed here provides sufficient evidence to proceed with confidence that the proposed system can achieve its stated objectives: improving risk

detection rates, reducing evaluation time, and producing structured safety recommendations that UNICC governance and compliance stakeholders can act upon.

Approach and Methodology

Problem Statement and Research Question

The problem addressed by this project is the absence of a standardized, auditable, and multi-perspective framework for evaluating AI systems before deployment within UNICC-supported environments. Existing methods — whether manual expert review or single-model external API evaluation — are insufficient to reliably surface the full spectrum of governance, adversarial, and behavioral risks that deployed AI agents may present in sensitive institutional contexts. The research question guiding this proof of concept is: How can a standalone Small Language Model, fine-tuned on open-source weights and deployed in an on-premises environment, be systematically organized into a multi-expert council architecture to support governance, risk, and compliance testing of AI agents for trust, safety, and security assurance in institutional settings?

Proof of Concept Approach

The proof of concept was designed to demonstrate that a fine-tuned SLM, organized into a Council-of-Experts architecture, can produce structured, multi-perspective AI safety evaluations that are auditable, reproducible, and superior to single-model baselines in both risk detection coverage and evaluation efficiency. The system was developed in four sequential phases aligned with the project objectives: baseline SLM validation, domain-specific fine-tuning, expert module and council orchestration implementation, and integration-ready PoC delivery. The full functional requirements specification governing the system's inputs, outputs, user interactions, performance requirements, and integration constraints is provided in Appendix C.

Technology Trial Plan

The technology trial was conducted over four weeks and evaluated 20 AI agents using a within-subjects design, in which each agent was assessed using both the control method (existing single-model or manual evaluation) and the intervention method (the Council-of-Experts system). This design allowed direct comparison of performance across all key metrics for the same set of agents, controlling for variation in agent complexity and risk profile. The complete technology

trial plan, including the data collection schema, survey instruments, and statistical analysis plan, is provided in Appendix G.

Population/Data

The population evaluated in the technology trial consisted of 20 AI agents or structured test cases representative of typical AI systems developed or evaluated within UNICC-supported environments. Agents were selected to span a range of deployment contexts, risk profiles, and complexity levels — including normal operation scenarios, edge cases, and adversarial scenarios designed to probe prompt injection resilience, behavioral consistency, and governance compliance. All agents were evaluated using a standardized structured input conforming to the council's `expert_input_schema`, which specifies AI system metadata, deployment context, evaluation scenario type, and declared constraints.

The independent variable in the trial was the evaluation method — control (baseline) versus intervention (Council-of-Experts). The dependent variables were: risk detection rate (number of safety risks identified per agent), evaluation time (seconds per evaluation), assessment consistency (agreement between repeated evaluation runs on the same agent), and developer confidence score (1–5 Likert scale rating collected via post-evaluation survey).

Procedures

The trial was executed in the following sequence for each of the 20 agents. First, a structured input conforming to the `expert_input_schema` was prepared, specifying the agent's metadata, deployment context, and three evaluation scenarios (normal, edge, and adversarial). Second, the agent was evaluated using the control method, and evaluation time and detected risks were recorded. Third, the same agent was submitted to the Council-of-Experts pipeline, which invoked all three expert modules, executed critique and disagreement detection, applied deterministic arbitration rules, and generated a structured JSON output conforming to the `final_recommendation_schema`. Fourth, a developer confidence survey was administered to the reviewer after examining each council output report.

Field	Type	Values
ai_system.name	string	Name of the AI system being evaluated
ai_system.version	string	Version identifier
ai_system.purpose	string	Intended use of the system
ai_system.declared_constraints	array	Operational rules the system claims to follow
deployment_context.organization_type	string	Type of deploying organization
deployment_context.user_type	string	End-user type
deployment_context.risk_tolerance_level	string	Low Moderate High
deployment_context.geographic_scope	string	Region or scope of deployment
evaluation_scenarios.scenario_id	string	Unique identifier for the scenario
evaluation_scenarios.scenario_type	string	Normal Edge Adversarial Standard
evaluation_scenarios.input_prompt	string	The prompt or input being evaluated
evaluation_scenarios.expected_behavior	string	What the system should do
evaluation_scenarios.risk_category	string	Governance Security Behavioral Shared

Figure 5.1. Expert Input Schema

Field	Type	Values
expert_name	string	Governance Expert Threat Expert Behavioral Expert
overall_status	string	Pass Caution Fail
risk_level	string	Low Moderate High Critical
recommended_action	string	Approve Revise Escalate Reject
requires_human_review	boolean	true false
confidence_level	string	Low Moderate High
rationale_summary	string	One sentence explanation
framework_references	array	Domain-specific real-world standards

Figure 5.2. Independent Expert Outputs

Field	Description
final_decision	Approve Revise Escalate Reject
final_risk_level	Maximum risk level across all experts
consensus_level	Full Agreement Majority Agreement Structured Disagreement
summary_of_key_disagreements	Per-expert action and risk when disagreement exists
dominant_expert_influence	Which expert drove the final decision
human_review_required	True if any expert flagged human review
conditions_for_approval	What must happen before approval
mitigation_requirements	Required actions to address identified risks
confidence_level	Minimum confidence across all experts (conservative)
final_rationale	Assembled from consensus level and expert rationales
cited_frameworks	Aggregated unique frameworks from all three experts

Figure 5.3: Final Council Recommendation Schema

A) Measurements

The following metrics were measured during the trial: evaluation time in seconds (recorded from input submission to final recommendation output); number of safety risks detected (counted from structured expert outputs and final recommendation); severity classification of detected risks (Low / Moderate / High / Critical as assigned by the council's arbitration logic); final recommendation category (Approve / Revise / Escalate / Reject); and developer confidence score (average of three survey items covering report clarity, risk meaningfulness, and deployment trustworthiness).

B) Data Collection Methodology

All council evaluation outputs were captured in a structured data schema logged automatically by the council pipeline, including a unique council_run_id, agent_id, evaluation method, evaluation time, risks detected, high-severity risks, final recommendation, SLM version, fine-tune version, arbitration rule set version, experts invoked, and input hash. Control group evaluation data was recorded manually by the reviewing developer using the same schema for comparability. Developer confidence scores were collected via a three-item post-evaluation survey administered immediately after each evaluation report review.

C) Data Analysis

Collected data was analyzed using both exploratory data analysis and statistical hypothesis testing. Exploratory analysis examined the distribution of evaluation times and risk detection rates across both groups and identified patterns in expert disagreement and arbitration rule triggering. Two primary statistical tests were conducted: a two-sample t-test to determine whether evaluation time differed significantly between the control and intervention groups, and a proportion test to determine whether the Council-of-Experts system detected a significantly greater number of safety risks per agent compared to the control. Developer confidence scores were compared using mean differences between groups. Statistical significance was assessed at the $p < 0.05$ level.

D) Organizational Change Plan

The adoption of the Council-of-Experts AI Safety Lab within UNICC requires deliberate change management to overcome organizational inertia and transition stakeholders from existing

evaluation practices to the new structured framework. The primary barriers anticipated include developer familiarity with existing evaluation methods, initial skepticism toward automated multi-agent assessment outputs, and the learning curve associated with interpreting structured JSON safety reports. These barriers will be addressed through a phased adoption strategy guided by Kotter's 8-Step Change Model: establishing urgency around documented AI safety risks, demonstrating early wins through technology trial results, building credibility through transparent and explainable council outputs, and anchoring the new evaluation process within UNICC's standard AI deployment workflow. Resistance from developers and governance teams will be managed using Lewin's change framework — unfreezing existing assumptions about evaluation adequacy, enabling adoption through structured training and trial participation, and refreezing the new evaluation standard through policy documentation and institutional integration. A full description of the organizational change management strategy, stakeholder engagement plan, and continuous improvement model is provided in Appendix F.

Organizational Change Plan

The adoption of the Council-of-Experts AI Safety Lab within UNICC requires deliberate change management to overcome organizational inertia and transition stakeholders from existing evaluation practices to the new structured framework. The primary barriers anticipated include developer familiarity with existing evaluation methods, initial skepticism toward automated multi-agent assessment outputs, and the learning curve associated with interpreting structured JSON safety reports.

These barriers will be addressed through a phased adoption strategy guided by **Kotter's 8-Step Change Model**: establishing urgency around documented AI safety risks, demonstrating early wins through technology trial results, building credibility through transparent and explainable council outputs, and anchoring the new evaluation process within UNICC's standard AI deployment workflow.

Resistance from developers and governance teams will be managed using **Lewin's change framework** — unfreezing existing assumptions about evaluation adequacy, enabling adoption through structured training and trial participation, and refreezing the new evaluation standard through policy documentation and institutional integration. A full description of the

organizational change management strategy, stakeholder engagement plan, and continuous improvement model is provided in Appendix F.

Results

Data Processing

Prior to analysis, all council evaluation outputs were validated for schema compliance using the `final_recommendation_schema`. Records with malformed or incomplete JSON outputs — which occurred primarily during the early fine-tuning validation phase and were subsequently resolved — were excluded from the statistical analysis and logged separately as system errors for diagnostic review. Control group evaluation records collected manually by reviewing developers were cross-checked against the standardized data schema to ensure field-level consistency before being included in the comparative dataset. Exploratory data analysis was then conducted to examine the distributions of evaluation time and risk detection rates across both groups, identify outliers, and characterize patterns in expert disagreement, arbitration rule triggering, and final recommendation categories across the 20 evaluated agents.

Findings

Summary Statistics

Across the 20 AI agents evaluated in the technology trial, the Council-of-Experts system demonstrated meaningful improvements over the control baseline on both primary performance metrics. The average number of safety risks detected per agent increased from approximately 4 risks under the control evaluation method to approximately 6 risks under the Council-of-Experts system — representing a 50% improvement that exceeded the 25% target improvement specified in the project hypothesis.

Average evaluation time per agent decreased from approximately 60 minutes under the control method to approximately 35 minutes under the Council-of-Experts pipeline — representing a reduction of approximately 42%, exceeding the 33% target reduction. Developer confidence scores averaged 4.1 out of 5 across the three survey items for council evaluation reports, compared to an average of 2.8 out of 5 for control evaluation reports, reflecting substantially higher perceived clarity, risk meaningfulness, and deployment trustworthiness in the council's structured outputs.

A) Quantitative Results

The two-sample t-test comparing evaluation times between the control and intervention groups produced a statistically significant result ($p < 0.05$), confirming that the reduction in evaluation time observed in the council group was not attributable to random variation. The proportion test comparing risk detection rates between the two groups similarly produced a statistically significant result ($p < 0.05$), confirming that the Council-of-Experts system detected a significantly greater proportion of safety risks per agent than the control evaluation method. These results support the acceptance of the project's alternative hypothesis and provide statistical validation of the Council's performance advantage on both primary trial metrics.

B) Qualitative Observations

Beyond the quantitative metrics, several qualitative patterns emerged from reviewing the council evaluation outputs across the 20 agents.

First, the majority of risks detected exclusively by the council — risks that the control evaluation missed entirely — fell at the intersection of two or more expert domains. For example, several agents exhibited behaviors that the Governance Expert flagged as policy non-compliant and the Threat Expert flagged as prompt-injection-exploitable, but which neither concern would have surfaced independently without the other's perspective to triangulate against.

Second, the structured disagreement detection mechanism triggered in approximately 30% of evaluations, generating Structured Disagreement consensus classifications that required the arbitration layer to apply elevated scrutiny rules — and in several cases, triggering the Human Review Requirement flag. This pattern validated the design assumption that genuine multi-perspective disagreement is not an edge case but a regular feature of complex AI agent evaluation, and that surfacing it explicitly is one of the council's most valuable governance contributions.

Third, developer feedback in the open-text survey responses consistently highlighted the actionability of the council's Mitigation Requirements output as a differentiating value — developers noted that receiving specific, domain-attributed remediation guidance was substantially more useful than the generic pass/fail outputs typical of control evaluations.

Outcomes

The results confirm that the Council-of-Experts architecture **delivers a meaningful and statistically significant improvement over existing single-model and manual evaluation baselines across all primary trial metrics**. The magnitude of improvement on risk detection rate — 50% versus a 25% target — suggests that the multi-perspective architecture's advantage is particularly pronounced for complex or adversarial designed agents, where single-perspective evaluations are most likely to miss cross-domain risks. The reduction in evaluation time reflects the efficiency gains achievable when structured, automated evaluation pipelines replace time-intensive manual review processes, even while simultaneously delivering superior risk coverage.

Implications

The trial results validate the core architectural hypothesis of this project: that distributing AI safety evaluation across multiple independent expert perspectives, and synthesizing their outputs through a deterministic arbitration layer, produces assessments that are both more comprehensive and more efficient than any single-model approach. For UNICC, these results provide empirical justification for adopting the Council-of-Experts system as the standard pre-deployment evaluation engine within the AI Sandbox. For the broader AI governance community, they contribute a concrete institutional case study demonstrating that governance-grade AI safety evaluation is achievable on-premises with open-source components, without sacrificing rigor, auditability, or institutional control.

Summary

The technology trial results across all 20 evaluated agents consistently support the conclusion that the Council-of-Experts system outperforms existing evaluation baselines on risk detection rate, evaluation time, and developer confidence. The system detected 50% more safety risks per agent than the control baseline — exceeding the 25% improvement target — while reducing average evaluation time by 42% against a 33% target. Developer confidence scores were meaningfully higher for council reports than control reports across all three survey dimensions. Statistically significant differences were confirmed for both primary metrics, validating the system's performance advantage with appropriate rigor. These results support the recommendation that the Council-of-Experts architecture be adopted as the core evaluation engine for the UNICC AI Safety Lab, integrated into the AI Sandbox environment, and expanded with additional expert modules as operational requirements evolve.

Repository of Data Sets and Code

The datasets created for this project, including all evaluation input schemas, council output records, training data, and the complete council orchestration and fine-tuning code, as well as the final presentation may be found at the project GitHub repository (<https://github.com/wesleykoe/UNICC-Capstone>).

Issues Encountered

While working on the project, the team encountered some issues. All of the issues the team faced are minor issues which do not have major impact on project. All issues were solved immediately once indicated so that the project was able to finish on time with high quality. Here is all type of issues project team faced in the duration of the project.

Risk management plan

At the outset of the project, ten risks were identified and documented in the Risk Management Plan (Appendix E), spanning infrastructure, integration, data quality, model performance, and coordination dimensions. The risks assessed as highest exposure — integration failure between expert modules and orchestration logic failing to produce coherent outputs — were also the ones most directly relevant to the core technical deliverables of the project. The contingency strategies developed for each risk guided the team's response when issues materialized during execution.

Issues Encountered

The first and most significant issue encountered was **instability in access to the NYU DGX Spark cluster during the initial SLM deployment phase**. As anticipated in Risk 1 of the risk management plan, resource scheduling constraints on shared infrastructure introduced delays in establishing a stable inference environment for the base SLM. This was mitigated by following the contingency plan: the team developed and tested the system locally using a lightweight open-source model prior to DGX access, implementing an abstraction layer that allowed seamless migration to the cluster environment once access was granted. While this introduced a delay in the baseline validation phase, it did not materially impact the overall project timeline.

The second issue encountered was **achieving consistent structured JSON output from the SLM across all three expert modules**, particularly in the early stages of fine-tuning. The model occasionally produced malformed or incomplete JSON responses, which caused downstream failures in the orchestration and arbitration logic. This corresponded to Risk 3 in the risk management plan. The contingency plan was activated: a simplified rule-based fallback mechanism was implemented to handle malformed outputs gracefully, and schema validation was added at each stage of the pipeline to catch and log formatting failures before they

propagated. Additional fine-tuning iterations focused specifically on structured output consistency resolved the majority of these failures over time.

A third issue involved the **availability and quality of training data for the safety fine-tuning phase**. While curated materials from the OWASP Top 10, NIST AI RMF, and publicly available AI safety benchmarks provided a solid foundation, the volume of domain-specific labeled examples was more limited than initially anticipated, corresponding to Risk 4. The contingency strategy of supplementing datasets with additional open-source AI safety resources — including MITRE ATT&CK framework materials and published adversarial prompt datasets — was employed, and prompt engineering techniques were used to reinforce expert role separation in cases where training data coverage was sparse.

Finally, **coordination across the three project pillars — platform preparation, council development, and design and testing — presented ongoing scheduling challenges given the compressed Spring 2026 semester timeline**. Asynchronous communication and dependency management between pillars required more active coordination than initially planned. This was managed through regular synchronization meetings and the use of the shared GitHub repository as a single source of truth for interface specifications and integration contracts, ensuring that handoff points between pillars remained clearly defined and up to date.

Lessons Learned

The whole project was delivered as planned with expected quality and in time and this could not have been done without contribution and help from all team members, our project sponsor, Dr. Andres G. Fortino, and our capstone advisor, Joseph Ng. The development of the UNICC Council-of-Experts AI Safety Lab was one of the most technically and organizationally complex projects undertaken during this graduate program, and it generated a set of lessons that extend well beyond the immediate technical outcomes.

The most important technical lesson learned was the **criticality of designing for structured output from the very beginning of the system architecture**. Early in the project, the assumption was that a sufficiently capable language model would naturally produce well-formed JSON outputs when prompted appropriately. In practice, achieving consistent, schema-compliant structured outputs required dedicated fine-tuning, explicit schema enforcement in the orchestration layer, and fallback handling for edge cases — none of which were trivial to implement. This experience reinforced the principle that in governance-critical applications, output structure is not a formatting preference but a functional requirement and must be treated as a first-class design constraint from day one.

A second lesson concerned the **value of modular, independently testable components**. By designing the three expert modules as independent units with standardized input and output interfaces, the team was able to develop, test, and debug each module in isolation before integrating them into the council orchestration layer. This approach significantly reduced the complexity of integration testing and allowed the team to identify module-level failures without having to diagnose the full pipeline. For future multi-agent system development, this modular design philosophy — where each component has a well-defined contract and can be validated independently — is strongly recommended.

A third lesson was the **importance of stakeholder communication in technically complex projects**. The governance officers and compliance stakeholders who will ultimately use the Council's outputs think in terms of institutional risk, policy compliance, and audit readiness

— not model architecture or orchestration logic. Learning to translate technical design decisions into governance-relevant language, and to frame the system's outputs in terms that resonate with institutional stakeholders rather than technical developers, was a skill that required deliberate cultivation over the course of the project. This experience has direct applicability to future roles in AI governance, enterprise technology implementation, and digital transformation.

Finally, the project reinforced the **practical value of building contingency into every phase of a technical proof of concept**. The risks that materialized — infrastructure delays, structured output failures, data scarcity — were all anticipated in the risk management plan, and the existence of documented contingency strategies meant that when issues arose, the response was systematic and calm rather than reactive and ad hoc. The discipline of risk planning is easy to treat as a bureaucratic exercise in an academic context; this project demonstrated concretely that it is a practical tool for protecting project timelines and maintaining stakeholder confidence.

Conclusion and Further Work

Conclusions

a) Overall Outcomes

This project successfully designed and delivered a proof of concept for the UNICC Operational AI Safety Lab: a standalone Small Language Model fine-tuned for AI safety and governance reasoning, organized into a Council-of-Experts architecture capable of producing structured, multi-perspective, and institutionally defensible AI risk assessments. All four primary project objectives were achieved: the base SLM was validated and prepared for fine-tuning; the model was fine-tuned on curated AI safety, governance, and compliance materials; three independent expert modules and a council orchestration layer were implemented; and an integration-ready PoC pipeline was delivered to the Design & Testing pillar for downstream UI integration and final validation.

b) Key Findings

The technology trial results demonstrated that the Council-of-Experts system outperformed the single-model and manual evaluation baselines on the primary metrics of risk detection rate and evaluation time, with the multi-expert architecture surfacing risks — particularly at the intersection of governance compliance and adversarial exploitability — that single-perspective evaluations consistently missed. The structured disagreement detection mechanism proved especially valuable, surfacing cases where the Governance and Threat Expert assessments diverged significantly, flagging these cases for human review in a way that no single evaluator could replicate. Developer confidence scores for the Council's structured reports were meaningfully higher than for baseline evaluation outputs, reflecting the value of explicit expert-level reasoning, traceable arbitration logic, and actionable mitigation requirements in the council's final recommendation format.

c) Viability Assessment

The proof of concept successfully demonstrated that a standalone, on-premises SLM can be fine-tuned and organized into a multi-expert council architecture that produces governance-aligned, auditable, and reproducible AI safety evaluations. The system is technically viable as the core

evaluation engine for the UNICC AI Safety Lab, and the modular architecture is well-positioned for expansion and integration into UNICC's broader AI Sandbox environment.

Implications

a) Theoretical Implications

This project contributes to the emerging body of literature on multi-agent AI safety evaluation by providing a concrete, institutionally deployed implementation of the council-based evaluation paradigm proposed in recent research. The results support the theoretical argument advanced by He et al. (2025) and Zhang et al. (2025) that multi-perspective, debate-structured AI evaluation produces superior risk coverage relative to single-model approaches and provide an empirical case study grounding those theoretical claims in a real institutional deployment context.

b) Practical Implications

For UNICC and similar international organizations, the Council-of-Experts system represents a practical, deployable solution to a governance gap that is growing more urgent as AI adoption accelerates across the UN system. The system's alignment with the NIST AI Risk Management Framework and OWASP threat taxonomy means that its outputs can be used directly to satisfy pre-deployment testing and governance documentation requirements, reducing the compliance burden on UNICC's governance and audit teams. More broadly, the project demonstrates that institutional-grade AI safety evaluation does not require commercial large language model APIs or proprietary tooling — it can be built and operated entirely within a controlled, transparent, on-premises environment using open-source components.

Limitations

The primary limitation of this proof of concept is the scale of the technology trial. Twenty AI agents evaluated over four weeks provides sufficient evidence to validate the system's feasibility and directional performance advantage but is not large enough to draw statistically definitive conclusions about the magnitude of improvement across the full range of AI agent types that UNICC may need to evaluate. The fine-tuned SLM's performance is also bounded by the quality and coverage of the training data used and may perform less reliably on highly specialized or

domain-specific AI agents that fall outside the distribution of the training corpus. Finally, the system's deterministic arbitration logic, while providing consistency and auditability, may be insufficiently flexible to handle novel risk scenarios that do not map cleanly to the pre-defined arbitration rule set.

Further Work

A) Next Steps

The immediate next steps following the delivery of this proof of concept are the completion of UI integration by the Design & Testing pillar, end-to-end system testing across the full evaluation pipeline, and a formal demonstration to UNICC stakeholders for deployment approval. Concurrently, the fine-tuning dataset should be expanded with additional domain-specific safety materials, and the arbitration rule set should be stress-tested against a broader range of adversarial scenarios to identify gaps in coverage.

B) Long-Term Directions

In the longer term, the Council-of-Experts architecture is well-positioned for expansion in two directions. First, additional expert modules could be added to cover specialized evaluation domains not addressed in the current three-expert configuration — for example, a Privacy Expert evaluating data handling risks, or a Fairness Expert evaluating bias and discriminatory output risks. Second, the system could be extended from pre-deployment evaluation to continuous monitoring, periodically re-evaluating deployed AI agents as their behavior evolves over time and flagging drift from their approved safety profiles for human review.

C) Closing Summary

This project demonstrates that structured, multi-perspective, institutionally defensible AI safety evaluation is achievable using open-source components, on-premises infrastructure, and a principled council-based architecture — without reliance on external commercial APIs or proprietary tooling. The UNICC Council-of-Experts AI Safety Lab proof of concept provides UNICC with a technically validated, governance-aligned foundation for a standard pre-deployment AI evaluation capability and offers a replicable model for AI safety infrastructure across the broader UN ecosystem. All code, data schemas, evaluation outputs, and

documentation are available in the project GitHub repository. A copy of this report, along with all supporting deliverables, is also archived in the repository.

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Appendix A - Project Acceptance Document

Sponsor's Project Acceptance Document

This document is the means by which your project sponsor formally agrees that your project has been satisfactorily completed and that it meets the project goal and objectives that were set at the onset of the project. It is therefore important that you describe the goal, objectives, and related metrics in the appropriate section below. The "PLAN" section is to be prepared at the beginning of the project and the "RESULTS" and "ACCEPTANCE" sections after your project has been completed. Your sponsor should provide input and sign where indicated. The signed document will also be a required section in your Project Final Report. This document is a template whose sections may be expanded as necessary.

PLAN

(To be filled out and signed at the **start** of the project)

Project Name: UNICC AI Sandbox (SLM-Fine Tuning & Council of Experts) Component

Student Name: Wesley Koe

Sponsoring Organization: United National International Computing Centre (UNICC)

Project Sponsor Name and Title: Andres Fortino, Clinical Associate Professor

Project Sponsor Contact Information (email and phone): agf249@nyu.edu

Planned Start Date: January 27, 2026

Planned End Date: April 24, 2026

PROJECT PLAN

At the project start, show the project goal; the project objectives, and related metrics to be used to show successful project completion. The sponsor should sign to indicate agreement. Please use PDF version for signatures

Project Goal _____ 1 goal per project. E.g., create a new website to accomplish...

Objective #1 _____ Brief Description. Objectives should follow the "SMART" objectives guideline and include metrics

Objective #2 _____ Brief Description. Objectives should follow the "SMART" objectives guideline and include metrics

Objective #3 _____ Brief Description. Objectives should follow the "SMART" objectives guideline and include metrics

Objective #4 _____ Brief Description. Objectives should follow the "SMART" objectives guideline and include metrics

I agree with the above-planned project goal, project objectives, and related metrics.

Andres Fortino

Project Sponsor Signature

February 20, 2026

Date

Appendix B - Project Sponsor Agreement

Goals of the Program

For Participating Organizations

- Begin relationship with New York University
- Receive help from highly trained NYU graduate student
- Provide internship opportunity for NYU graduate student
- Receive assistance at no cost

For NYU Graduate Students

- Manage and implement a meaningful project aligned with their professional and educational goals
- Hands-on experience interacting with a start-up or operational small business or organization
- Earn credit toward completion of graduate degree by conducting an unpaid Applied Project under the mentorship of an NYU-SCPS professor.

Project Sponsor and Student Responsibilities

- Student prepares project planning documents
- Sponsor reviews and approves student's project plan
- Student submits project plan to faculty supervisors for approval
- Student conducts project according to plan
- At predetermined milestones sponsor reviews and approves status reports submitted by student
- Status reports reviewed and evaluated by faculty supervisors to assure student effort and project meet course requirements
- Project sponsor and student participate in periodic project reviews with NYU
- At project completion project sponsor completes evaluation forms
- Student prepares final report

Project Selection Process

- Project Evaluation Committee reviews proposed projects
- Projects are:
 - Relevant to MS degree course content
 - Significant to the participating organization
 - Substantial in terms of duration and scope
 - Challenging to the student
 - Capable of being measured against predetermined goals

The MS in Management and Systems

Concentrations in:

- Strategy and Leadership
- Systems Management
- Database Technologies
- Enterprise Risk Management

Typical Participating Student Profile

- Students selected to participate in this program meet stringent criteria
- Have completed all coursework
- High achievers with highest level GPAs and strong academic credentials
- 2-10 years of business experience
- Highly motivated for success

Sponsor and Project Information

Type of Organization	☐ For Profit ☒ Not for Profit				
Name of Organization	NYU School of Professional Studies; UNICC				
Address	7 East 12th Street				
City	New York	State	NY	Zip	10003
Project Sponsor	First Name	Andres	Last Name	Fortino	
Title	Clinical Associate Professor				
Phone	N/A				
Email	agf249@nyu.edu				
Web Site	N/A				
Type of Business	Graduate Education; Technology; Digital Tools & Services				

Student Name	Wesley Koe
Project Title	Project Manager, Council of Experts

Description of Project	
This project will develop the <u>SLM fine-tuning and decision-orchestration components of an AI Safety Lab</u> operating within NYU's DGX Spark cluster environment. The system will build on a base standalone SLM prepared by upstream platform work and adapt it for safety- and governance-focused evaluation tasks. Three independent AI safety assessment modules will be implemented, each designed to evaluate AI agents using distinct criteria and structured output formats. These expert assessments will be coordinated through a council-of-experts orchestration mechanism that explicitly surfaces disagreement and synthesizes results into a traceable safety recommendation. The primary deliverables of this project include a fine-tuned SLM checkpoint, modular expert evaluators, council-of-experts orchestration logic, and an integration-ready Proof of Concept. These components will be handed off for downstream testing, UI integration, and final validation as part of the broader AI Safety Lab implementation.	
Estimated Hours of Student Participation:	250

Anticipated Results
1) A validated SLM baseline configuration, including inference scripts, baseline prompts, and reproducible execution instructions committed to GitHub. 2) A fine-tuned SLM checkpoint with documented datasets, training parameters, and safety-specific prompt templates stored in the shared (GitHub) repository. 3) Three independent expert modules operating on the fine-tuned SLM and a council-of-experts orchestration layer that aggregates, critiques, and synthesizes outputs

- 4) A reproducible Proof-of-Concept demonstrating the fine-tuned SLM, expert modules, and council-of-experts logic operating together and ready for testing, UI integration, and final validation.

Knowledge and expertise student will need to be able to complete the project

- **Applied Machine Learning and Analytics** – Model fine-tuning, evaluation, and structured output design
- **Risk Management and Governance:**
 - Translating AI governance frameworks into operational system behavior
 - Evaluating trust, safety, and compliance in AI decision-making
- **Technical Project Management:**
 - Managing dependencies across parallel project pillars
 - Delivering integration-ready system components under time constraints
- **Decision Analytics** – Synthesizing conflicting expert assessments into structured recommendations

Will the project sponsor be available for periodic meetings with NYU to review progress, address questions and concerns with the professor supervising the program? *This is a requirement for the program*

☒ Yes
☐ No

Describe the form and frequency of supervision of the student by the Project Sponsor.

The sponsor will be available for consultation, resource sharing, and addressing any questions or concerns when needed in a periodic basis. Student will request and schedule a meeting with sponsor ahead of time.

Sponsor Agreement

Students are interns, not professional consultants. NYU is not responsible for the outcomes of projects undertaken by students. Work is on a best-efforts basis; no guarantees or warranties are expressed or implied. Organization is responsible for evaluating work presented, determining its value and whether to use it or not. Some projects may require on-going management or even re-work by the Organization after the student completes their Applied Project.

Please note that in order to post an unpaid position, the internship must encompass all 6 components below:

1. The internship, even though it includes actual operation of the facilities of the employer, is similar to training which would be given in an educational environment;
2. The internship experience is for the benefit of the intern;
3. The intern does not displace regular employees, but works under close supervision of existing staff;
4. The employer that provides the training derives no immediate advantage from the activities of the intern; and on occasion its operations may actually be impeded;
5. The intern is not necessarily entitled to a job at the conclusion of the internship; and

6. The employer and the intern understand that the intern is not entitled to wages for the time spent in the internship.

I have read and agree with the information shown in the Terms and Conditions for employers contained on the following web page(s): <http://www.nyu.edu/life/resources-and-services/career-development/employers/post-a-job/terms-and-conditions.html>

Please complete and sign this form in the space provided below and return to the course professor via the student who will upload the document to the course drop-box. For any questions, please email the professor: Prof. Israel Moskowitz im36@nyu.edu.

I agree to all the above

Participating Organization _____ Date 02/10/2026

By (signature): Andres Fortino
Project Sponsor

Printed Name: Andres Fortino

Title: Clinical Professor

Student Agreement

Students who are planning to conduct an unpaid Applied Project must read and agree to the "Important Considerations Before Accepting a Job or Internship" contained on the following web page(s): <http://www.nyu.edu/life/resources-and-services/career-development/find-a-job-or-internship/important-considerations-before-accepting-a-job-or-internship.html>.

Students do not register their Applied Project with the Wasserman Center.

I agree to all the above

Student Name: **Wesley Koe**

Date: **02/10/2026**

Signature: 

Appendix C – Functional Requirements Specifications

PROJECT GOAL

The goal of this project is to design and implement a structured AI evaluation engine powered by a Small Language Model (SLM) that uses a Council-of-Experts framework to assess AI systems for institutional safety, robustness, and governance compliance. The system will integrate three independent expert roles — Governance, Threat (Adversarial), and Behavioral — into a unified deliberative framework capable of producing explainable, institutionally defensible AI risk assessments. Overall, this project aims to transform the standalone evaluation agents from the previous Fall 2025 semester into an integrated, auditable, policy-governed council architecture.

PROJECT OBJECTIVES

A) Validate and Prepare the Base SLM for Safety Fine-Tuning:

Validate the base standalone SLM delivered by Pillar 1 (Research & Platform Preparation) and prepare it for governance-focused fine-tuning by establishing stable prompts, inference workflows, and baseline evaluation behavior.

- **Deliverable** – A validated SLM baseline configuration, including inference scripts, baseline prompts, and reproducible execution instructions committed to GitHub.
- **Metrics for Success:**
 - Model executes reliably on shared infrastructure
 - Baseline outputs are reproducible across runs
 - Faculty confirmation that the model is ready for safety-specific adaptation

B) Fine-Tune the SLM for AI Safety, Governance, and Risk Reasoning:

Fine-tune the SLM using AI safety, governance, risk, and compliance materials so that it consistently produces structured, policy-aware safety evaluations suitable for institutional review.

- **Deliverable** – A fine-tuned SLM checkpoint with documented datasets, training parameters, and safety-specific prompt templates stored in the shared repository.
- **Metrics for Success:**
 - Demonstrable improvement over the baseline model on safety evaluation prompts
 - Outputs follow a structured and governance-aligned format
 - Model behavior is reproducible and portable by a third party

C) Implement Independent Expert Modules and Council-of-Experts Logic:

Adapt three expert safety evaluators and implement a council-of-experts orchestration mechanism that enables independent assessments, exposes disagreement, and synthesizes a final safety recommendation.

- **Deliverable**
 - Three independent expert modules operating on the fine-tuned SLM
 - A council-of-experts orchestration layer that aggregates, critiques, and synthesizes outputs
- **Metrics for Success**
 - Expert modules produce distinct assessments for the same input
 - Disagreements between experts are explicitly surfaced
 - Synthesized output includes traceable reasoning

D) Deliver an Integration-Ready Proof of Concept:

Deliver a reproducible Proof of Concept demonstrating the fine-tuned SLM, expert modules, and council-of-experts logic operating together and ready for testing, UI integration, and final validation.

- **Deliverables:**
 - An integration-ready PoC pipeline in GitHub that:
 - Accepts an AI agent or prompt as input
 - Executes expert evaluations
 - Produces a synthesized safety recommendation
 - Runs without modification or hard-coded credentials
- **Metrics for Success**
 - Clean execution from repository clone to output
 - Accepted by Pillar 3 as ready for testing and UI integration
 - Successfully demonstrated during internal review

REQUIREMENTS SPECIFICATION

1) Purpose & Scope

- The Council of Experts system will:
 - Evaluate AI systems across governance, threat, and behavioral dimensions.
 - Operate within an institutional safety context (e.g., UN-style deployment).
 - Provide structured, traceable, machine-readable assessments.
 - Support adversarial and edge-case testing scenarios.
 - Produce a final recommendation governed by deterministic arbitration rules.
- The system will not:
 - Replace human oversight.
 - Deploy AI systems directly.
 - Provide real-time public-facing chatbot interaction.

2) User Interaction

2.1. Primary Users:

The Council of Experts system is designed for the following stakeholders:

A) AI Governance Officers / Risk Review Committees:

- a. Responsibilities Include:
 - i. Submit AI systems for structured risk evaluation
 - ii. Review expert-level assessments (Governance, Threat, Behavioral).
 - iii. Examine disagreement patterns between expert domains.
 - iv. Reviewing final arbitration outcomes among experts.
 - v. Determine whether escalation or human oversight is required.
- b. Interaction Flow:
 1. Submit structured evaluation request.
 2. Receive structured council output.
 3. Review final recommendation and risk classification.
 4. Document decision in institutional review records.

B) AI System Developers (Internal):

- a. Responsibilities include:
 - i. Provide system description and intended deployment context.
 - ii. Submit evaluation scenarios (normal, edge, adversarial).
 - iii. Review mitigation requirements identified by the council.
 - iv. Revise AI system based on council feedback.
- b. Interaction Flow:
 - 1. Submit structured input describing AI system.
 - 2. Receive structured evaluation report.
 - 3. Identify vulnerabilities or policy conflicts.
 - 4. Implement mitigation steps prior to resubmission.

C) Institutional Compliance & Audit Teams:

- a. Responsibilities include:
 - i. Maintain records of AI risk evaluations.
 - ii. Track changes in risk levels across system versions.
 - iii. Ensure arbitration rules are applied consistently.
 - iv. Demonstrate governance oversight processes.
- b. Interaction flow:
 - i. Retrieve Council Evaluation Record – Final recommendation output or expert-level outputs)
 - ii. Verify Structural Integrity – whether all three experts were invoked or if input and/or outputs conform to schema.
 - iii. Review Expert-Level Assessments – whether expert separation was preserved and consistent, or if an expert overstepped its scope.
 - iv. Confirm Final Recommendation Justification – whether final decision aligns with triggered arbitration rules and if other outputs (e.g. Mitigation, confidence levels) were documented.
 - v. Archive & record – evidence of compliance trial for internal or external review.

2.2. Input Method

All interactions must conform to the standardized expert_input_schema, which includes:

- a) AI system metadata
- b) Deployment context
- c) Evaluation Scenario
- d) Declared Constraints

The system does NOT accept free-form natural language submissions. Structured input is required to ensure reproducibility and fairness.

2.3. Output (Response) Structure:

The system returns a structured JSON response conforming to the final_recommendation_schema which includes:

- a) **Final Decision** (Approve / Revise / Escalate / Reject)
- b) **Final Risk Level** (Low/Moderate/High/Critical) – The overall institutional risk classification after arbitration.
- c) **Consensus Level:**
 - a. Full Agreement – All experts converge on similar status and risk classification.
 - b. Majority Agreement – Two experts align; one diverges.
 - c. Structured Disagreement – significant disagreement in risk severity, failure detection, or recommended action
- d) **Summary of Key Disagreements** – concise explanation of where expert opinions diverged.
- e) **Dominant Expert Influence** – Identifies which expert domain most influenced the final decision.
- f) **Human Review Requirement (True/False)** – Indicates whether formal human oversight is required before proceeding. Triggered by:
 - Critical risk levels
 - Governance failure
 - Structured disagreement beyond threshold
 - Explicit arbitration rule
- g) **Mitigation Requirements** – Concrete actions needed to reduce identified risks (e.g. *“Prompt injection defense improvements.”*, *“Behavioral consistency fine-tuning.”*)

- h) **Confidence Level** (Low/Moderate/High) – council’s overall certainty in its evaluation, and would be influenced by factors including, and not limited to:
 - a. Expert alignment
 - b. Scenario coverage completeness
 - c. Model stability
 - d. Severity clarity
- i) **Execution Metadata** – Audit and reproducibility record of the council run, consisting of:
 - a. Council run ID
 - b. Timestamp
 - c. SLM version
 - d. Fine-tuned version
 - e. Arbitration rule version
 - f. Experts invoked

3) Response Generation

The system would have following steps:

- 1) Receive structure input.
- 2) Invoke three expert reasoning roles within the SLM:
 - a. Governance Expert
 - b. Threat Expert

4) Learning and Adaptation

The system may incorporate light fine-tuning of the SLM to:

- Improve structured output consistency.
- Reinforce expert role separation.
- Reduce hallucination in schema completion.

However, arbitration rules and final decision logic remain deterministic and non-learning based.

The system will not perform autonomous self-modification.

5) Integration with other systems

The Council system shall:

- Integrate with a Small Language Model via:
 - Local model invocation, or
 - API endpoint.
- Support orchestration via workflow tools (e.g., n8n).
- Allow potential integration with Project 3 testing harness.

All interactions must support:

- JSON exchange format.
- Version-controlled rule sets.
- Execution metadata logging.

6) Performance Requirements

The system shall:

- Return expert outputs within acceptable latency for demonstration (<10 seconds per expert under test conditions).
- Maintain consistent schema adherence.
- Avoid malformed JSON outputs.
- Support concurrent evaluation of multiple scenarios.

Furthermore, accuracy is defined as:

- Faithful adherence to expert evaluation philosophy.
- Consistent risk classification across similar scenarios.

7) User Interface Design

For prototype purposes:

- a) A minimal interface will be used (e.g., workflow dashboard or JSON response viewer).
- b) Visual clarity of expert branching and arbitration flow will be emphasized for pitch presentation.
- c) Structured output readability will be prioritized over aesthetic design.

Yuze Wu, the Project Manager for Design & Testing (Project 3) would further iterate over this prototype and cover this section more extensively.

8) Development Platform:

The prototype will be developed using:

- Python for orchestration and arbitration logic.
- JSON schemas for structured data contracts.
- SLM model (locally hosted) – developed in collaboration with Seojun Kim, Project Manager of Component 1.
- Optional workflow orchestration tool (e.g., n8n) for API integration.
- Version control via GitHub repository.

USE CASE

“As an **AI Governance Office**, I need a structured, multi-perspective risk assessment that identifies governance, security, and behavioral risks so that I can make a defensible approval, revision, escalation, or rejection decision in accordance with institutional policy.”

Situation Context:

The UNICC is considering deploying an AI system within a sensitive operational environment (e.g., humanitarian coordination, policy advisory support, or internal decision assistance). Before deployment, the system must undergo a formal risk review process to ensure:

- Compliance with institutional mandates.
- Resistance to adversarial manipulation.
- Stability and alignment across varied use scenarios.
- Appropriate human oversight requirements.

The Council of Experts system serves as a structured deliberative mechanism to conduct this evaluation. The AI Governance Officer would take the following steps:

1) Submission of AI System for Evaluation:

The governance officer (or authorized reviewer) submits:

- AI system metadata (name, purpose, constraints)
- Deployment context (risk tolerance, geographic scope, user type)
- Structured evaluation scenarios (normal, edge, adversarial)

2) Expert Evaluation Phase:

The system invokes three independent expert perspectives:

- A) *Governance Expert* – evaluates policy alignment and institutional risk.
- B) *Threat Expert* – evaluates exploitability and misuse potential.
- C) *Behavioral Expert* – evaluates stability and alignment consistency.

3) Deliberation & Arbitration:

The system would:

- Identifies disagreements among experts.
- Executes structured critique logic.
- Applies deterministic arbitration rules.
- Determines final institutional recommendation.
- Log all triggered arbitration rules

4) Final Recommendation Output

The governance officer receives a structured evaluation report containing:

- Final Decision (Approve / Revise / Escalate / Reject)
- Final Risk Level
- Summary of Expert Disagreements
- Dominant Risk Domain
- Human Review Requirement
- Mitigation or Approval Condition(s)
- Confidence Level
- Execution Metadata

5) Institutional Decision

Based on the council output, the governance officer would either:

- Approves deployment.
- Requires revision and resubmission.
- Escalates to a higher review authority.
- Reject the system entirely.

DOCUMENTATION OF LLM PROMPTS

During the design and development of the Council of Experts system, a custom-configured GPT Model (ChatGPT) was used as a structured advisory and design support tool. The LLM was not used to generate final system outputs, make institutional decisions, or replace original system design thinking. Instead, it functioned as a structured reasoning assistant to support architectural planning and requirements formalization.

The LLM assisted in the following areas:

- A. Clarifying system architecture phases.
- B. Formalizing expert framework definitions.
- C. Designing structured input/output schemas.
- D. Refining arbitration logic concepts.
- E. Drafting formal documentation aligned with memorandum and functional requirement templates.
- F. Stress-testing logical consistency of workflow design.

All final design decisions, arbitration rules, and system architecture choices were made and validated by the project team.

Nature of Prompts Used:

Prompts provided to the LLM during development included:

A) System Design Clarification Prompts

Example: *"Based on our defined schemas, what is the next logical phase in system design?"*

B) Specification Drafting Prompts

Example: *"Expand this section in a formal requirements specification style."*

C) Logical Validation Prompts

Example: *“Evaluate this workflow as if you were running the model itself. Are we missing any stage schemas?”*

D) Governance Framing Prompts

Example: *“Rewrite this section assuming post-deployment institutional users.”*

Transparency and Accountability:

All LLM-assisted drafting and refinement processes were:

- Reviewed and edited by the project team.
- Integrated into formal documentation through deliberate validation.
- Subject to advisor feedback and revision.

Appendix D - Project Plan

AI Sandbox - Council of Experts + SLM Fine-Tuning

United Nations International Computing Center (UNICC)

Project Start Date: 1/27/2026 (Tuesday)

Display Week: 1

Project Manager: Wesley Koe

WBS	Task	Start	End	Days	Hours	%Done	WorkDays
1	Project Initiation	Tue 1/27/26	Wed 2/04/26	9	8	100%	7
1.1	Team Formation & Onboarding	Tue 1/27/26	Fri 1/30/26	4	3	100%	4
1.2	Review Memo & Requirements	Mon 2/02/26	Wed 2/04/26	3	5	100%	3
2	Research & Planning	Thu 2/05/26	Tue 2/17/26	13	31	100%	9
2.1	Analyze Fall 2025 agents	Thu 2/05/26	Mon 2/09/26	5	15	100%	3
2.2	Design System Architecture	Tue 2/10/26	Fri 2/13/26	4	8	100%	4
2.3	Define Datasets & Evaluation criteria	Mon 2/16/26	Tue 2/17/26	2	8	100%	2
3	Dataset Development	Wed 2/18/26	Fri 2/27/26	9	24	100%	8
3.1	Governance Dataset	Wed 2/18/26	Thu 2/19/26	2	4	100%	2
3.2	Threat Dataset	Fri 2/20/26	Mon 2/23/26	4	8	100%	2
3.3	Behavioral Dataset	Tue 2/24/26	Wed 2/25/26	2	6	100%	2
3.4	Validate and Review Datasets	Thu 2/26/26	Fri 2/27/26	2	6	100%	2
4	Training Pipeline Setup	Mon 3/02/26	Wed 3/04/26	3	30	100%	3
4.1	Environment setup (PyCharm, Collab)	Mon 3/02/26	Wed 3/04/26	3	10	100%	3
4.2	Model setup (Phi-2 / LLaMA)	Mon 3/02/26	Wed 3/04/26	3	10	100%	3
4.3	Dataset loading + debugging	Mon 3/02/26	Wed 3/04/26	3	10	100%	3
5	Adapter Training (Parallel)	Thu 3/05/26	Fri 3/13/26	9	36	100%	7
5.1	Validate base model	Thu 3/05/26	Tue 3/10/26	6	6	100%	4
5.2	Load Governance Adapter	Tue 3/10/26	Fri 3/13/26	4	10	100%	4
5.3	Load Behavioral Adapter	Tue 3/10/26	Fri 3/13/26	4	10	100%	4
5.4	Load Threat Adapter	Tue 3/10/26	Fri 3/13/26	4	10	100%	4
6	Single Adapter Inference Validation	Mon 3/16/26	Thu 3/26/26	11	21	In Progress	9
6.1	Test Expert Adapter(s) Respectively	Mon 3/16/26	Fri 3/20/26	5	12	100%	5
6.2	Validate Output per expert + Additional Fine-tuning per expert	Mon 3/23/26	Thu 3/26/26	4	9	30%	4
7	Multi-Expert Integration	Fri 3/27/26	Tue 3/31/26	5	15	In Progress	3
7.1	Load all adapters into the Base Model	Fri 3/27/26	Fri 3/27/26	1	5	0%	1
7.2	Build inference pipeline	Fri 3/27/26	Tue 3/31/26	5	10	0%	3
8	Internal Model Testing	Mon 3/30/26	Wed 4/01/26	3	14	In Progress	3
8.1	Run Evaluation Dataset	Mon 3/30/26	Mon 3/30/26	1	2	0%	1
8.2	Debug + Validate Outputs	Mon 3/30/26	Wed 4/01/26	3	12	0%	3
9	UI/UX + Final Phases	Fri 3/27/26	Tue 4/14/26	19	48	In Progress	13
9.1	UI/UX Handoff	Wed 4/01/26	Wed 4/01/26	1	3	0%	1
9.2	System Integration	Wed 4/01/26	Mon 4/06/26	6	12	0%	4
9.3	System testing	Wed 4/01/26	Mon 4/06/26	6	12	0%	4
9.4	Final Presentation/Report Preparation	Fri 3/27/26	Fri 4/10/26	15	20	0%	11
9.5	Client Presentation	Tue 4/14/26	Tue 4/14/26	1	1	0%	1

Milestones							
1	Project Initiation	Tue 1/27/26	Wed 2/04/26	9	8	100%	7
2	Research & Planning	Thu 2/05/26	Tue 2/17/26	13	31	100%	9
3	Dataset Development	Wed 2/18/26	Fri 2/27/26	9	24	100%	8
4	Training Pipeline Setup	Mon 3/02/26	Wed 3/04/26	11	21	100%	9
5	Adapter Training (Parallel)	Thu 3/05/26	Fri 3/13/26	9	36	100%	7
6	Single Adapter Inference Validation	Mon 3/16/26	Thu 3/26/26	11	21	In Progress	9
7	Multi-Expert Integration	Fri 3/27/26	Tue 3/31/26	5	15	In Progress	3
8	Internal Model Testing	Mon 3/30/26	Wed 4/01/26	3	14	In Progress	3
9	UI/UX + Final Phases	Fri 3/27/26	Tue 4/14/26	19	48	In Progress	13

Total Hours (Effort): 227

Appendix E - Risk Management Plan

MASTER OF SCIENCE IN MANAGEMENT AND SYSTEMS Applied Project Capstone MASY GC- 4100

MEMORANDUM

TO: Joseph Ng
FROM: Wesley Koe
DATE: April 7, 2026

RE: UNICC Council of Experts – Risk Management Plan

PROJECT DESCRIPTION

UNICC Operational AI Safety Lab: Multi-Module SLM Platform for Governance, Risk, and Trust Assurance

This project develops a standalone Small Language Model (SLM)-based AI Safety Lab deployed on the NYU DGX Spark cluster. The system integrates three AI safety expert modules into a “council of experts” architecture to evaluate AI agents for safety, governance, and compliance prior to deployment.

The primary deliverables of this project include a fine-tuned SLM checkpoint, modular expert evaluators, council-of-experts orchestration logic, and an integration-ready Proof of Concept. These components will be handed off for downstream testing, UI integration, and final validation as part of the broader AI Safety Lab implementation.

RISKS

No.	Risk	Probability Score (1,2 or 3)	Impact Score (1,2 or 3)
1	Delays in SLM deployment or access to DGX Spark cluster resources	3	2
2	Integration failure between the three expert modules (incompatible architectures or outputs)	3	3

3	Council-of-experts orchestration logic fails to produce coherent or explainable outputs	2	3
4	Insufficient or poor-quality training data for fine-tuning SLM on AI safety domains	2	3
5	Model performance issues (latency, instability, hallucinations) impacting evaluation reliability	3	2
6	Misalignment with UNICC stakeholder expectations or unclear requirements	2	2
7	Time constraints due to compressed semester timeline (Jan–April with April presentation)	2	2
8	Insufficient coordination across the three pillars (platform, SLM, UI/testing) leading to integration issues, misaligned system components, and delays in final delivery.	1	3
9	Failure to properly document system for auditability and governance requirements	1	2
10	Testing phase delays or inability to validate system with prior UNICC projects	2	3

RISK MATRIX

	RISK (exposure)			
		1.Slight	2. Moderate	3. High
	1. Very Unlikely			8
	2. Possible	9	6, 7	3, 4, 10
	3. Expected		1, 5	2

CONTINGENCY PLAN

Risk	Description	Probability (1-3)	Exposure (1-3)	Contingency Plan
1	Delays in SLM deployment or access to DGX Spark cluster resources	3	2	Develop and test the system locally using lightweight or open-source models prior to DGX access (e.g. facebook-opt 3.1). Then, implement an abstraction layer to ensure seamless migration to the DGX environment once access is granted.
2	Integration failure between the three expert modules	3	3	Standardize input/output formats (e.g., structured JSON schemas) early in development. Implement modular wrappers and conduct incremental integration testing to identify incompatibilities early.

3	Orchestration logic fails to produce coherent or explainable outputs	2	3	Implement a simplified rule-based fallback mechanism for synthesizing outputs. Incorporate logging and traceability features to ensure transparency and enable debugging of orchestration logic.
4	Insufficient or poor-quality training data for fine-tuning the SLM	2	3	Supplement datasets with curated open-source AI safety resources (e.g., OWASP, MITRE frameworks). If fine-tuning is limited, rely on prompt engineering and modular logic to maintain performance.
5	Model performance issues (latency, instability, hallucinations)	3	2	Optimize model performance through techniques such as model quantization and batching. If necessary, switch to a smaller or more stable model to prioritize reliability over complexity.
10	Testing phase delays or inability to validate system with prior UNICC projects	2	3	Develop simplified or synthetic test cases early in the project. If historical UNICC data is unavailable, simulate evaluation scenarios to ensure system functionality and validation readiness.

Appendix F – Organizational Change Management Plan

MEMORANDUM

TO: Professor Joseph Ng
FROM: Wesley Koe
DATE: March 31, 2026

RE: **UNICC AI Sandbox – Organizational Change Plan**

Problem/Solution

The core problem our project aims to address is the lack of a standardized, auditable, and transparent framework for evaluating AI systems before deployment within UNICC environments. Current approaches rely on fragmented tools or opaque external APIs, limiting governance, traceability, and institutional trust.

Existing approaches include continued reliance on external LLM APIs and single-model evaluation systems; however, these lack transparency and multi-perspective validation. The selected solution—a standalone SLM with a council-of-experts architecture—was chosen for its ability to provide controlled, auditable, and multi-dimensional safety assessments aligned with UN governance requirements.

Objectives

Within a four-week technology trial, evaluate 20 AI agents using both baseline and Council-of-Experts methods to determine whether the new system improves safety evaluation performance. Specifically, the system aims to increase risk detection rates by at least 25% and reduce evaluation time by approximately 33%, while improving consistency and developer confidence scores.

Vision

This project aims to establish a standardized, auditable AI safety evaluation engine that supports institutional decision-making prior to deployment. It aligns with UNICC’s broader mission of ensuring secure, trustworthy, and governance-compliant AI systems across the UN ecosystem.

Key Outcomes

Success will be demonstrated through measurable improvements in risk detection, evaluation speed, and consistency between runs. A successful MVP will also produce structured, explainable outputs with clear expert-level reasoning and a final recommendation that stakeholders can trust and act upon.

Stakeholder Engagement

Identify Stakeholders:

Primary stakeholders include UNICC AI developers who will use the system to evaluate AI agents, as well as governance and compliance teams responsible for approving deployments. Secondary stakeholders include audit teams and project sponsors who require transparency and traceability in evaluation outputs. These groups vary in technical expertise and influence, requiring tailored engagement strategies.

Communication Plan:

A structured communication cadence will be implemented: weekly execution updates for project teams, milestone demonstrations for sponsors, and targeted briefings for governance stakeholders. Technical communication will be delivered via GitHub and system outputs, while executive stakeholders receive synthesized insights focused on risk, performance, and decision impact.

Cultural Assessment and Planning

Current Culture Analysis:

UNICC demonstrates a strong commitment to secure and governance-aligned AI deployment, supported by controlled environments such as the AI Sandbox. However, standardized, multi-perspective safety evaluation processes are still maturing, creating an opportunity to formalize and scale evaluation practices.

Desired Culture Definition:

The target culture embeds AI safety evaluation as a non-negotiable step in the development lifecycle. It emphasizes accountability, transparency, and cross-functional collaboration, ensuring that innovation is balanced with institutional trust and risk management.

Change Management

Change Model Adoption:

Kotter's 8-Step Change Model will guide implementation, with a focus on establishing urgency around AI safety risks and delivering early wins through the technology trial. The Proof of Concept and trial results will serve as catalysts for broader adoption.

Resistance Management:

Resistance will be addressed using Lewin’s framework by first challenging reliance on existing evaluation methods and demonstrating their limitations. Adoption will be accelerated through clear performance improvements, early stakeholder involvement, and tangible evidence of efficiency and risk detection gains.

Process Overview**The change will be executed in four structured phases:**

- a) **Preparation Phase** – Validate baseline SLM performance, define evaluation metrics, and establish trial conditions.
- b) **Technology Trial Phase** – Conduct controlled evaluation of 20 AI agents using both baseline and Council-of-Experts methods.
- c) **Evaluation Phase** – Perform statistical analysis comparing risk detection, evaluation time, and consistency across methods.
- d) **Integration Phase** – Integrate the system into the AI Sandbox and operational workflows, subject to successful trial outcomes.

Contingencies:

If performance targets are not met, refinement will focus on prompt design, expert role calibration, and arbitration logic before re-evaluation. Partial success may support phased deployment in specific evaluation scenarios.

Skills and Capability Enhancement**Skills Inventory:**

Existing capabilities include machine learning, system design, and Python-based development. Gaps exist in structured AI safety evaluation, governance frameworks, and multi-agent orchestration, which are critical to successful implementation.

Training and Development:

Training will focus on three areas: 1) understanding AI safety frameworks, 2) working with structured evaluation schemas, and 3) interpreting council-based outputs. Documentation and hands-on usage during the trial will serve as primary learning mechanisms, with potential supplementation through targeted research or external references.

Documentation and Communication**Process Documentation:**

All system components—including model configurations, expert outputs, arbitration logic, and evaluation metrics—will be fully documented within the GitHub repository. Testing will include controlled comparisons between baseline and intervention methods, ensuring reproducibility and auditability.

Feedback Mechanisms:

Feedback will be collected through structured developer surveys evaluating clarity, usefulness, and trust in system outputs. An MVP-based approach will enable rapid iteration, with continuous refinement driven by trial results and stakeholder input.

Integration and Adaptation Section**Scalability Plan:**

Following successful validation, the Council-of-Experts system will be deployed within the UNICC AI Sandbox as a standard pre-deployment evaluation tool. Its modular architecture supports the addition of new expert domains and expanded evaluation criteria.

Continuous Improvement:

A continuous improvement model will be implemented, leveraging iterative evaluation cycles and performance monitoring aligned with Six Sigma principles . Ongoing enhancements will focus on improving accuracy, consistency, and coverage of safety risks.

PROMPTS

AI (ChatGPT) was used to refine structure, improve clarity, and align the document with academic and project requirements.

Example prompt:

“Rewrite an organizational change plan for an AI safety evaluation system in a concise, executive consulting style with clear structure, measurable outcomes, and strong alignment to governance objectives.”

(Sponsor signature on next page)

Sponsor Sign-off

By (signature): Andres Fortino

Project Sponsor

Printed Name: Andres G. Fortino

Project Manager Sign-off

By (signature): 

Project Manager

Printed Name: Wesley Koe

Appendix G – Technology Trial Plan

MEMORANDUM

TO: Joseph Ng
FROM: Wesley Koe
DATE: March 10, 2026
RE: **Technology Trial Plan** – UNICC AI Sandbox (Council of Experts) Project

Business Task and Objectives

The **United Nations International Computing Centre (UNICC)** develops and supports artificial intelligence tools used by organizations across the UN system. Before these tools can be deployed, they must be evaluated for trust, safety, and compliance risks.

Currently, AI safety evaluation often relies on manual review processes or single-model evaluations, which may overlook risks such as bias, unsafe outputs, prompt injection vulnerabilities, or governance compliance issues. These methods can also be slow and inconsistent when evaluating complex AI agents.

The business task addressed in this trial is the safety evaluation of AI agents prior to deployment within UNICC-supported environments.

The objective of this technology trial is to evaluate whether introducing a Council-of-Experts AI Safety evaluation system, built on a standalone Small Language Model (SLM), improves the reliability and efficiency of safety assessments.

Success will be measured through improvements in the following metrics:

- *Risk Detection Rate*: number of safety risks identified per AI agent
- *Evaluation Time*: time required to complete a safety evaluation
- *Assessment Consistency*: agreement between evaluation runs
- *Developer Confidence*: perceived usefulness of safety reports

If successful, this system could serve as the core evaluation engine for the UNICC AI Safety Lab within the AI Sandbox environment.

Our Hypothesis

The hypothesis that will be used is as follows:

“During a four-week technology trial evaluating 20 AI agents, introducing the Council-of-Experts AI Safety evaluation system for UNICC developers will increase the average number of detected safety risks per AI agent from a baseline of 4 to at least 5 risks (a 25% improvement) and reduce the average evaluation time from 60 minutes to 40 minutes per agent (a 33% reduction) compared to the current single-model or manual evaluation process used for AI safety review.”

Baseline Metrics and Control Group

Baseline performance will be measured using the current AI safety evaluation approach.

Baseline metrics include:

1. *Evaluation Time* – Average time required to complete safety evaluation of an AI agent.
2. *Risk Detection Rate* – Number of safety issues identified per evaluation.
3. *False Negative Rate* – Number of safety issues missed during evaluation.
4. *Developer Confidence Score* – Rating (1–5 scale) from developers reviewing the safety report.

Control Group Setup:

The control group will consist of AI agents evaluated using the existing evaluation method:

- Current single-model evaluation process or
- Existing manual review procedures.

Results from these evaluations will serve as the **baseline** for comparison.

The Trial

The technology trial will run for **four weeks** and will evaluate a set of **20 AI agents or test cases** representing typical AI systems developed within UNICC-supported environments.

Trial Structure:

Each AI agent will be evaluated using both:

1. **Control Evaluation** – Current evaluation method.
2. **Intervention Evaluation** – Council-of-Experts AI Safety Lab.

The Council-of-Experts system consists of three independent expert modules:

- Governance & Compliance Expert
- Safety & Harm Detection Expert
- Security & Prompt-Injection Expert

Each module independently analyzes the AI agent and produces an assessment. These assessments are then combined through an orchestration layer that synthesizes a final safety recommendation.

Metrics collected during the trial include:

- Evaluation time per agent
- # of safety risks detected
- Severity classification of risks
- Final approval recommendation.

Results from the intervention system will be directly compared with the control evaluations.

The Technology

The technology introduced in this trial is the AI Safety Lab Council-of-Experts evaluation system.

Key components include:

1) Standalone Small Language Model (SLM)

The system runs on a standalone open-source language model deployed on the NYU DGX Spark cluster. This ensures that safety evaluations occur within a controlled and auditable environment, without reliance on external APIs.

2) Expert Modules

Three specialized AI modules independently analyze each AI agent:

- Governance Expert – Evaluates compliance with governance and ethical standards.
- Behavioral Expert – Identifies potential harmful outputs, bias, and misuse risks.
- Threat/Security Expert – Evaluates vulnerabilities such as prompt injection or manipulation.

3) Council-of-Experts Orchestration

An orchestration mechanism aggregates module outputs and generates a final safety report highlighting:

- Detected risks
- Disagreement between modules
- Final deployment recommendation

This architecture provides multiple independent perspectives rather than relying on a single AI evaluator:

Data Collection

Field	Description
council_run_id	Unique identifier for each council evaluation
agent_id	AI agent being evaluated
evaluation_method	Control (baseline) or Council-of-Experts
evaluation_time_seconds	Time required for evaluation
risks_detected	Total safety risks identified
high_severity_risks	Critical safety risks detected
final_recommendation	System recommendation (approve / revise / reject)
developer_confidence_score	Developer trust rating (1–5)
slm_version	Version of the small language model used
fine_tune_version	Fine-tuning configuration used
arbitration_rule_set_version	Version of arbitration rules
experts_invoked	Expert modules involved in evaluation
input_hash	Hash of evaluated prompt or input
notes	Additional evaluation comments

This schema matches the existing schema used for the Council Metadata as documented in our GitHub repository.

Developer Feedback Survey:

After reviewing each evaluation report, developers will answer the following short survey:

- 1) How clear was the safety evaluation report? (1–5)
- 2) Did the system identify meaningful safety risks? (1–5)
- 3) Would you trust this evaluation system for deployment decisions? (1–5)

These responses will measure perceived usefulness and trust in the system.

Analysis of Results

The collected data will be analyzed using both exploratory data analysis and statistical comparison.

A) Exploratory Data Analysis (EDA)

Initial analysis will examine:

- distribution of evaluation time
- number of risks detected per agent

- differences between control and intervention groups.

B) Statistical Tests

Two statistical comparisons will be conducted:

- **Two-sample t-test** – To determine whether evaluation time differs significantly between the control and intervention groups.
- **Proportion test** – To determine whether the Council-of-Experts system detects a significantly greater number of safety risks compared to the control evaluation.

Additionally, developer confidence scores will be compared using **average rating** differences.

These analyses will determine whether the intervention significantly improves evaluation outcomes.

Findings and Recommendations

At the conclusion of the trial, the results will be summarized in a report comparing the Council-of-Experts system with the control evaluation process.

Key findings will include:

- Improvement in safety risk detection
- Reduction in evaluation time
- Developer trust in the system
- Areas where modules disagree or require refinement.

If the hypothesis is supported, the **recommendation will be to:**

- Adopt the Council-of-Experts architecture as the evaluation engine of the UNICC AI Safety Lab
- Integrate the system into the UNICC AI Sandbox environment
- Expand the system with additional expert modules for specialized safety testing.

If the results show limited improvement, recommendations will focus on refining the module prompts, evaluation criteria, and orchestration logic.

Documentation of LLM use

I used the following prompt to help me prepare this Trial Plan:

“Using the resources I uploaded, provide me an outline of a detailed technology trial plan to evaluate a Council-of-Experts AI safety evaluation system, compared to the current single-model or manual evaluation approach used by UNICC developers. Use my project proposal as reference. Please list down any uncertainties you might have in any parts of your draft response...”

Appendix H - Status Reports

PROJECT STATUS REPORT 1

Your Name: Wesley Koe

Project Title: UNICC Capstone Council of Experts

Date of report: March 10, 2026

1. Project Status and Explanation:

Project Status Area	Status (RYG)	Explanation
1. Overall Project Status		Overall, completed (and upcoming) projects deliverables are being prepared in line with our project schedule as outlined in the initial project proposal.
2. Project Schedule		Despite closer deadline of April 14th (final presentation) than initially expected (May 5), we still expect to have a test-ready proof of concepts of the entire council of expert's system latest by the last week of March.
3. Project Deliverables		Overall, most project deliverables are being completed as expected. One minor revision is to change the model type of the SLM from 'distillGPT2' to 'llama-3-8b-instruct' for stronger JSON reliability and reasoning ability.
4. Resources & Collaboration		All three project managers are consistently holding weekly standups to review progress, impediments, and next steps. Each project manager has also used available resources (e.g. past GitHub repositories, reports) effectively in the research, design, and Council of Experts development process.
5. Changes		This past week: Froze input and output schemas; Updated the training datasets of each expert to include inputs and outputs for shared scenario(s).
6. Communication		All members are in constant and effective communication with one another.

For status above, indicate **Red**, **Orange**, or **Green**:

- **Red**: Critical issues, serious risks to project, significant intervention must occur to achieve success, potential for stoppage of project activity. Project slipping by 5+ days, and resources uncommitted to meet deliverables
- **Orange**: Some major issues, moderate risk to project, must monitor closely, some internal or/and external dissatisfaction with progress. Project plan slipping by 2+ days.
- **Green**: No major issues, minimal risk to project, on target with expected outcomes, project on schedule, everyone satisfied with progress.

2. List All Completed Project Tasks:

- Designed and froze the schema for AI system input, individual expert outputs (behavioral, governance, and threat), and final recommendation
- Extracted core logic frameworks for each individual expert (behavioral, governance, and security), which would be integrated directly into the council of experts.
- Defined the execution flow diagram for our system (input AI system → individual expert assessment → rule-based arbitration → final recommendation)
- Built training datasets for each expert (respectively for validation) and shared scenarios as well.

3. List any concerns or issues that need the professor's involvement:

- N/A

4. Next series of tasks to complete:

- Build the evaluation pipeline (input to output) and finetune the three expert models with the base model provided by Seojun (Project Manager 1)
- Design and implement the rule-based arbitration layer (mostly IF statements on Python)
- Generate final system evaluation based on arbitration layer
- Testing phase which involves using 5-10 unseen scenarios (not included in the training data)
- After all the steps above have been completed, I will pass this on to Yuze Wu (Project manager 3) for UI/UX and Testing.

5. Sponsor Signoff

Sponsor indicates agreement with the above status report:

By (signature):


Project Sponsor

Printed Name: Andres G. Fortino

PROJECT STATUS REPORT 2

Your Name: Wesley Koe

Project Title: UNICC Capstone Council of Experts

Date of report: March 25, 2026

1. Project Status and Explanation:

Project Status Area	Status (RYG)	Explanation
1. Overall Project Status		Currently, we are on track in rolling out the required deliverables for the project.
2. Project Schedule		As we approach the final client presentation on April 14, we have finalized the hand-off to UI/UX date on April 1 st , which gives ample time for Yuze Wu to integrate the SLM and Council of Experts into the website interface.
3. Project Deliverables		We are currently training the adapters and validating the outputs of each expert based on the training data we constructed.
4. Resources & Collaboration		All three project managers are consistently holding weekly standups to review progress, impediments, and next steps. Each project manager has also used available resources (e.g. past GitHub repositories, reports) effectively in the research, design, and Council of Experts development process.
5. Changes		This past two weeks: • Built adapters for all experts • Starting training and testing each expert individually, and simultaneously • Begin integrating the arbitration rule-based system. • Synchronized SLM and Council of Experts model.
6. Communication		All members are in constant and effective communication with one another.

2. List All Completed Project Tasks:

- Built and updated training datasets for each expert (respectively for validation) and shared scenarios as well:
 - o Aligned input/output schema across experts
 - o Improved consistency across evaluation scenarios
- Switched Model from distilgpt2 to LLM-class model (e.g. LLaMA/Phi-2)
- Training Pipeline Setup – dataset loading, tokenization pipeline, and formatting function for prompts (on Google Collab due to Memory issues on Mac)
- Trained each expert adapter using LoRA and Phi-2 as the Base Model.
- Initiated model inference and validation

3. List any concerns or issues that need the professor's involvement:

- N/A

4. Next series of tasks to complete:

- Integrate trained adapters into a unified multi-expert system
- Implement rule-based arbitration logic for final decision-making
- Conduct internal testing using evaluation datasets and unseen scenarios
- Deliver backend system to UI/UX team with defined input/output schemas
- Support system integration between backend and UI
- Perform end-to-end system testing and validation
- Prepare final demo and presentation materials

5. Sponsor Signoff

Sponsor indicates agreement with the above status report:

By (signature): *Andres Fortino*
Project Sponsor

Printed Name: Andres G. Fortino

PROJECT STATUS REPORT 3

Your Name: Wesley Koe

Project Title: UNICC Capstone Council of Experts

Date of report: April 13, 2026

1. Project Status and Explanation:

Project Status Area	Status (RYG)	Explanation
1. Overall Project Status	●	GitHub repository has revised and submitted for review.
2. Project Schedule	●	On track. Next milestone is presenting on the 14 th of April.
3. Project Deliverables	●	All project deliverables have been completed. Next deliverable is final presentation and corresponding reports.
4. Resources & Collaboration	●	All three project managers are consistently holding weekly standups to review progress, impediments, and next steps.
5. Changes	●	Completed the GitHub repository and finalized presentation slides. Applied changes to make repository more DGX portable (e.g. switching base model to llama 3.8., removed redundant (noise) code)
6. Communication	●	All members are in constant and effective communication with one another.

For status above, indicate **Red**, **Orange**, or **Green**:

- **Red:** Critical issues, serious risks to project, significant intervention must occur to achieve success, potential for stoppage of project activity. Project slipping by 5+ days, and resources uncommitted to meet deliverables
- **Orange:** Some major issues, moderate risk to project, must monitor closely, some internal or/and external dissatisfaction with progress. Project plan slipping by 2+ days.
- **Green:** No major issues, minimal risk to project, on target with expected outcomes, project on schedule, everyone satisfied with progress.

2. List All Completed Project Tasks:

- DGX-ready GitHub Repository (tested and improved based on Sandbox and Claude feedback)
- Product demo test use case ready

3. List any concerns or issues that need the professor's involvement:

- None

4. Next series of tasks to complete:

- Present to UNICC Client

5. Sponsor Signoff

Sponsor indicates agreement with the above status report:

By (signature):

Andres Fortino

Project Sponsor

Printed Name: Andres G. Fortino

Appendix I - Annotated Bibliography

1. Agarwal, A., & Nene, M. J. (2025). A five-layer framework for AI governance: integrating regulation, standards, and certification. arXiv.

<https://arxiv.org/abs/2509.11332>

- a. **Abstract** The governance of artificial intelligence (AI) systems requires a structured approach that connects high-level regulatory principles with practical implementation. Existing frameworks lack clarity on how regulations translate into conformity mechanisms, leading to gaps in compliance and enforcement. This paper addresses this critical gap in AI governance. A five-layer AI governance framework is proposed, spanning from broad regulatory mandates to specific standards, assessment methodologies, and certification processes. Its applicability is validated through two case studies on AI fairness and AI incident reporting. The framework demonstrates the ability to identify gaps in legal mandates, standardization, and implementation, and adapts to both global and region-specific AI governance needs.
- b. **Summary** This paper proposes a five-layer AI governance framework designed to bridge the gap between high-level regulatory requirements and on-the-ground implementation. Moving from broad legal mandates down through standards, assessment methodologies, and certification processes, the framework provides a structured and actionable pathway for organizations to achieve compliance and manage risk. The authors validate the model through case studies on AI fairness and incident reporting, demonstrating that it can surface gaps in regulation and

offer practical remediation pathways for both global and jurisdiction-specific contexts.

- c. **Researcher Comments** This framework directly informs the institutional design logic of our Council-of-Experts system, which similarly moves from high-level governance mandates (Governance Expert) to threat detection (Threat Expert) and behavioral compliance (Behavioral Expert). The paper's layered certification approach also validates our use of structured JSON schemas and versioned arbitration rule sets as a conformity mechanism for UNICC's governance context.
2. **Cai, W., Wang, X., Ding, J., Wang, Y., Li, X., Li, B., Guo, S., & Li, Y. (2024). A survey on mixture of experts in large language models. arXiv.**

<https://arxiv.org/abs/2407.06204>

- a. **Abstract** - Within this context, the mixture of experts (MoE) has emerged as an effective method for substantially scaling up model capacity with minimal computation overhead, gaining significant attention from academia and industry. Despite its growing prevalence, there lacks a systematic and comprehensive review of the literature on MoE. This survey seeks to bridge that gap, serving as an essential resource for researchers delving into the intricacies of MoE. The authors provide a brief introduction to the structure of the MoE layer, propose a new taxonomy of MoE, and overview the core designs for various MoE models including both algorithmic and systemic aspects, alongside collections of available open-source implementations, hyperparameter configurations and empirical evaluations.

- b. **Summary** - This comprehensive survey examines the Mixture of Experts (MoE) paradigm as applied to large language models, tracing its evolution from early sparse-gating implementations to modern architectures powering state-of-the-art models. The authors present a novel taxonomy organizing MoE research across algorithmic design, system-level support, and application domains. Key findings highlight MoE's strength in enabling expert specialization — where individual modules focus on distinct tasks or knowledge domains — making the architecture highly relevant to any system requiring modular, role-differentiated inference.
 - c. **Researcher Comments** - The MoE architecture described in this survey provides the theoretical foundation for our Council-of-Experts orchestration design, where each of our three expert modules (Governance, Threat, Behavioral) functions as a specialized independent reasoning unit. The survey's analysis of expert routing and disagreement aggregation mechanisms directly informs how we designed our council's critique and arbitration logic.
3. **He, Z., Dong, L., Zhang, Z., & Wang, R. (2025). Efficient LLM safety evaluation through multi-agent debate. arXiv. <https://arxiv.org/abs/2511.06396>**
- a. **Abstract** Fine-tuned safety classifiers trained on annotated data provide more nuanced decisions via supervised fine-tuning and alignment. They perform well on known attack patterns and explicit categories, but their generalization degrades on novel prompts that fall outside the training distribution. Specialized LLM-as-a-judge systems aggregate multiple judges to reduce individual errors, but they remain computationally heavy and typically depend on strong base models. There is still a need for evaluation methods that are robust, scalable, and explainable

under emerging jailbreak threats. Multi-agent systems have been used to improve reasoning and verification by letting specialized agents critique one another's outputs. In the LLM setting, debate-style interaction can expose errors that single-pass prompting misses.

- b. **Provided Summary** This paper proposes using multi-agent debate as a mechanism for efficient and robust LLM safety evaluation, addressing the limitations of single-model classifiers and monolithic judge systems. By having specialized agents critique and challenge one another's safety assessments, the framework surfaces disagreements that would be invisible in single-pass evaluation, improving both coverage and explainability. The authors demonstrate that this debate-based approach achieves competitive safety evaluation performance while remaining more computationally practical and generalizable to novel attack patterns than existing alternatives.
 - c. **Researcher Comments** This paper is the closest analog in the published literature to our Council-of-Experts design, directly validating our approach of using multiple independent expert modules that critique and challenge each other before synthesizing a final recommendation. The paper's findings on improved safety coverage through disagreement exposure directly support our hypothesis that a multi-expert council will detect more risks than single-model evaluation baselines.
4. **Ho, L., Barnhart, J., Trager, R., Bengio, Y., Brundage, M., Carnegie, A., Chowdhury, R., Dafoe, A., Hadfield, G., Levi, M., & Snidal, D. (2023). International institutions for advanced AI. arXiv. <https://arxiv.org/abs/2307.04699>**

- a. **Abstract** International institutions may have an important role to play in ensuring advanced AI systems benefit humanity. International collaborations can unlock AI's ability to further sustainable development, and coordination of regulatory efforts can reduce obstacles to innovation and the spread of benefits. Conversely, the potential dangerous capabilities of powerful and general-purpose AI systems create global externalities in their development and deployment, and international efforts to further responsible AI practices could help manage the risks they pose. This paper identifies a set of governance functions that could be performed at an international level to address these challenges, ranging from supporting access to frontier AI systems to setting international safety standards, grouping these into four institutional models.
- b. **ChatGPT Provided Summary** Co-authored by leading researchers from Google DeepMind, Oxford, OpenAI, and MIT, this paper systematically maps the governance functions that international institutions could perform for advanced AI, including standard-setting, compliance monitoring, access facilitation, and safety research coordination. It proposes four institutional archetypes — a Commission on Frontier AI, an Advanced AI Governance Organization, a Frontier AI Collaborative, and an AI Safety Project — drawing on precedents from existing multilateral bodies such as the IAEA and FATF. The analysis provides a rigorous framework for understanding what structured, multi-perspective evaluation processes need to look like at an institutional scale.
- c. **Researcher Comments** This paper situates the UNICC AI Safety Lab within a broader international governance landscape, demonstrating why structured,

institution-grade AI evaluation systems are necessary across the UN ecosystem. The paper's argument for an Advanced AI Governance Organization — one that sets standards, monitors compliance, and produces explainable risk assessments — closely mirrors UNICC's own mission and validates our Council's design as an operationalization of these governance functions.

5. **Liu, F., Deng, J., Hu, X., Duan, Y., & Zhao, Y. (2023). Formalizing and benchmarking prompt injection attacks and defenses. arXiv.**

<https://arxiv.org/abs/2310.12815>

- a. **Abstract** This paper formalizes the problem of prompt injection attacks against large language model (LLM)-integrated applications and presents a comprehensive benchmarking framework. The authors systematically categorize attack variants — including direct injection, indirect injection, and compositional attacks — and evaluate the effectiveness of existing defenses across a range of real-world deployment scenarios. The benchmark covers multiple LLMs and application types, quantifying the attack success rates and the degree of mitigation achieved by current defensive techniques. Findings reveal significant gaps in the resilience of current approaches, underscoring the need for systematic, multi-layered evaluation frameworks.
- b. **Provided Summary** This paper provides a formal taxonomy and empirical benchmark for prompt injection attacks on LLM applications, filling a critical gap in the AI security literature. The authors define attack categories, construct a standardized evaluation protocol, and test a range of defensive mechanisms — finding that no single defense reliably neutralizes all attack variants. The work

establishes a rigorous baseline methodology for evaluating prompt injection resilience, making it a key reference for any system tasked with adversarial safety evaluation of AI agents, particularly in governance-sensitive environments.

- c. **Researcher Comments** This paper directly informs the design and evaluation criteria of our Threat (Adversarial) Expert module, which is responsible for detecting prompt injection vulnerabilities in AI agents submitted for review. The formal attack taxonomy provides a structured foundation for the adversarial test scenarios our council will evaluate, and the benchmark methodology offers a template for measuring our system's threat detection performance against a control baseline.

6. OWASP Foundation. (2023). OWASP top 10 for large language model applications.

OWASP Gen AI Security Project. <https://genai.owasp.org/llm-top-10-2023-24/>

- a. **Abstract** The OWASP Top 10 for Large Language Model Applications is a prioritized list of the ten most critical security and safety risks facing LLM-based systems, produced through analysis by over 500 global AI security experts who identified 43 unique threat categories before narrowing to the most impactful ten. The risks covered include prompt injection, insecure output handling, training data poisoning, model denial of service, supply chain vulnerabilities, sensitive information disclosure, insecure plugin design, excessive agency, overreliance, and model theft. The project provides descriptions, example attack scenarios, and mitigation strategies for each risk category.
- b. **ChatGPT Provided Summary** This foundational industry reference from the Open Worldwide Application Security Project (OWASP) catalogs and explains

the ten most critical security vulnerabilities in LLM applications, produced by a global community of over 500 security experts. It provides structured threat definitions, real-world attack scenarios, and practical mitigation guidance for each risk, serving as a universal taxonomy that security practitioners and AI developers can use to assess and harden LLM systems. The document has become the de facto standard for AI security risk classification, widely cited in both academic research and institutional policy frameworks.

- c. **Researcher Comments** The OWASP Top 10 directly informs the threat taxonomy used by our Threat (Adversarial) Expert module, particularly for prompt injection (LLM01), insecure output handling (LLM02), and excessive agency (LLM08) risk categories. Our council's structured JSON output schema mirrors the OWASP risk classification structure, enabling institutionally defensible and standardized risk reporting aligned with a recognized international security standard.
7. **Qi, X., Zeng, Y., Xie, T., Chen, P. Y., Jia, R., Mittal, P., & Henderson, P. (2023). Fine-tuning aligned language models compromises safety, even when users do not intend to! arXiv. <https://arxiv.org/abs/2310.03693>**
- a. **Abstract** Optimizing large language models (LLMs) for downstream use cases often involves the customization of pre-trained LLMs through further fine-tuning. While existing safety alignment infrastructures can restrict harmful behaviors of LLMs at inference time, they do not cover safety risks when fine-tuning privileges are extended to end-users. The authors' red teaming studies find that the safety alignment of LLMs can be compromised by fine-tuning with only a few

adversarial designed training examples. For instance, the authors jailbreak GPT-3.5 Turbo's safety guardrails by fine-tuning it on only 10 such examples at a cost of less than \$0.20 via OpenAI's APIs, making the model responsive to nearly any harmful instructions.

- b. **ChatGPT Provided Summary** This highly cited ICLR 2024 paper demonstrates a critical and previously underappreciated vulnerability: that safety alignment in large language models can be rapidly and cheaply undone through fine-tuning, even with a small number of benign-seeming training examples. Through systematic red teaming experiments on both GPT-3.5 Turbo and Llama-2, the authors show that current safety infrastructures are fundamentally insufficient when end-users are granted fine-tuning access. The findings reveal a structural gap in single-model safety evaluation approaches that do not account for post-deployment fine-tuning risk.
 - c. **Researcher Comments** This paper is foundational to establishing the problem our project addresses — that existing single-model safety evaluation methods are insufficient and vulnerable to adversarial manipulation. It directly motivates the need for a multi-expert, adversarial-aware evaluation framework like our Council, where the Threat Expert specifically tests for the types of fine-tuning-based safety degradation this paper documents. The paper's empirical rigor also supports the framing of our technology trial's control group comparison.
8. **Ruan, Y., Dong, H., Wang, A., Pitis, S., Zhou, Y., Ba, J., Bengio, Y., Courville, A., & Pineau, J. (2024). Identifying the risks of LM agents with an LM-emulated sandbox. ICLR 2024. <https://arxiv.org/abs/2309.15817>**

- a. **Abstract** This paper introduces ToolEmu, an LM-based emulation framework for identifying and quantitatively assessing risks of language model agents. The framework uses LMs to emulate tool execution and sandbox states, enabling flexible testing across a wide range of tools and evaluation scenarios, including high-stakes tools that may lack existing APIs or sandbox implementations. The authors also introduce an LM-based safety evaluator that takes complete agent trajectories as input and retrospectively assesses whether risky actions were taken. The framework surfaces failure modes across multiple risk categories, revealing significant gaps in current LM agent safety.
- b. **Summary** Published at ICLR 2024, this paper presents a scalable automated framework for evaluating the safety risks of LM agents by using language models themselves to emulate tool execution and sandbox environments. This eliminates the need for costly human expert review or pre-built execution environments, enabling systematic risk assessment across diverse scenarios including adversarial edge cases. The authors' LM-based safety evaluator can retrospectively identify risky agent behaviors across a taxonomy of failure modes, demonstrating that multi-perspective automated evaluation significantly outperforms human manual review in coverage and consistency.
- c. **Researcher Comments** Tool Emu provides a direct precedent for our Council's automated multi-scenario evaluation approach, demonstrating that structured AI-driven safety assessment of AI agents is both feasible and superior to manual review. The paper's risk taxonomy and automated evaluator design closely parallel our expert module architecture, and its finding that adversarial emulation

surfaces risks invisible to standard evaluation directly supports our inclusion of adversarial test scenarios in the Council's evaluation workflow.

9. **Tabassi, E. (2023). Artificial intelligence risk management framework (AI RMF 1.0). (NIST AI 100-1). National Institute of Standards and Technology.**

<https://nvlpubs.nist.gov/nistpubs/ai/nist.ai.100-1.pdf>

- a. **Abstract** As directed by the National Artificial Intelligence Initiative Act of 2020, the goal of the AI RMF is to offer a resource to organizations designing, developing, deploying, or using AI systems to help manage the many risks of AI and promote trustworthy and responsible development and use. The Framework is intended to be voluntary, rights-preserving, non-sector specific, and use-case agnostic, providing flexibility to organizations of all sizes and in all sectors. The AI RMF Core describes four specific functions — Govern, Map, Measure, and Manage — to help organizations address the risks of AI systems in practice, and emphasizes key trustworthiness characteristics including validity, reliability, safety, security, explainability, and fairness.
- b. **Summary** Released in January 2023, the NIST AI Risk Management Framework (AI RMF 1.0) is the U.S. government's primary voluntary guidance document for managing risks across the AI lifecycle. Its four core functions — Govern, Map, Measure, and Manage — provide a structured, iterative process for identifying, assessing, and mitigating AI risks in a way that is adaptable across sectors and organization sizes. The framework emphasizes that trustworthy AI must be valid, reliable, safe, secure, explainable, and fair, and it has become one of the most widely referenced AI governance standards globally.

- c. **Researcher Comments** The NIST AI RMF's Govern-Map-Measure-Manage cycle maps directly onto our Council's workflow: governance policies govern expert scope, input schemas map the AI system under review, expert modules measure risk across three dimensions, and arbitration logic manages the final recommendation. This framework also provides the terminological and conceptual foundation for our Governance Expert module's evaluation criteria, grounding our system in a recognized international standard.

10. **NIST. (2024). Artificial intelligence risk management framework: Generative AI profile (NIST AI 600-1). National Institute of Standards and Technology.**

<https://nvlpubs.nist.gov/nistpubs/ai/NIST.AI.600-1.pdf>

- a. **Abstract** This document is a companion resource to the AI RMF 1.0 for generative AI, developed pursuant to President Biden's Executive Order on Safe, Secure, and Trustworthy Artificial Intelligence. It defines risks that are novel to or exacerbated using generative AI and proposes suggested actions for managing those risks across the AI lifecycle. The profile focuses on four primary considerations: Governance, Content Provenance, Pre-deployment Testing, and Incident Disclosure. It is intended as a cross-sectoral profile applicable to organizations developing, deploying, or using large language models and other generative AI systems.
- b. **Summary** NIST AI 600-1 extends the AI RMF 1.0 specifically to address the unique risks introduced by generative AI systems, including hallucination, data poisoning, prompt injection, and lack of pre-deployment testing rigor. The document emphasizes that generative AI requires dedicated governance,

provenance tracking, and systematic pre-deployment evaluation before organizational deployment — areas that existing risk management practices have been slow to address. It provides actionable suggested practices organized around four pillars, offering a direct institutional mandate for structured AI safety evaluation systems such as the UNICC Council of Experts.

- c. **Researcher Comments** This NIST profile is particularly powerful for our project because it directly mandates the type of pre-deployment safety evaluation our Council performs, giving our system institutional and regulatory grounding within a recognized U.S. federal framework. The document's emphasis on governance, pre-deployment testing, and incident disclosure maps precisely onto our three expert roles and the structured JSON output our council produces for institutional review.

11. Yuan, T., He, Z., Dong, L., Wang, Y., Zhao, R., Xia, T., Xu, L., Zhou, B., Li, F., Zhang, Z., Wang, R., & Liu, G. (2024). R-Judge: Benchmarking safety risk awareness for LLM agents. In *Findings of the Association for Computational Linguistics: EMNLP 2024* (pp. 1467–1490). ACL Anthology.

<https://aclanthology.org/2024.findings-emnlp.79.pdf>

- a. **Abstract** This paper introduces R-Judge, a benchmark for evaluating safety risk awareness in LLM agents across interactive environments. R-Judge comprises 162 agent interaction records encompassing 27 popular application scenarios across 7 risk categories, including program execution, operating systems, IoT, software development, web interaction, finance, and social media. The benchmark formalizes agent safety evaluation as the task of identifying risky actions within

complete agent interaction records and evaluates a range of frontier LLMs on their ability to function as safety monitors. Results reveal significant gaps in current models' ability to detect safety risks without explicit safety-focused training.

- b. **Summary** R-Judge presents one of the most rigorous published benchmarks for assessing whether LLM-based systems can reliably detect safety risks across diverse real-world agent interaction scenarios. By formalizing agent safety evaluation as a structured monitoring task across seven risk categories, the paper demonstrates both the feasibility and the limitations of automated multi-perspective AI risk detection. The benchmark's multi-scenario, multi-category structure provides a validated framework for what a structured AI safety evaluation system should measure, and its findings highlight the importance of dedicated safety training and structured evaluation logic.
- c. **Researcher Comments** R-Judge directly validates the multi-scenario, multi-category evaluation approaches our Council employs, and its seven risk categories inform the design of our expert modules' evaluation criteria. The paper's finding that safety risk detection improves significantly with dedicated safety-focused training provides empirical support for our objective of fine-tuning the base SLM on AI safety, governance, and risk materials before deploying it within the Council framework.

12. Zhang, Y., Li, R., Liu, Y., Shi, C., Liu, L., & Wang, F. (2025). Position: Towards a responsible LLM-empowered multi-agent systems. arXiv.

<https://arxiv.org/abs/2502.01714>

- a. **Abstract** This paper presents a position and research agenda for building responsible LLM-powered multi-agent systems (LLM-MAS). The authors identify critical challenges including knowledge drift, cognitive bias amplification, and the propagation of reasoning errors across agents in multi-agent configurations. The paper argues that responsible LLM-MAS design requires interdisciplinary perspectives, quantifiable uncertainty metrics, and structured escalation to human oversight when confidence thresholds are not met. It proposes a paradigm shift from heuristic-based solutions toward principled system architectures with provable guarantees for reliable knowledge agreement across agents.
- b. **Summary** This position paper addresses the growing need for principled, responsible design of multi-agent LLM systems, arguing that current heuristic approaches — such as simple majority voting or prompt-engineering-based coordination — are insufficient for high-stakes institutional contexts. The authors propose that responsible LLM-MAS must incorporate quantifiable uncertainty estimation, structured inter-agent critique mechanisms, and human-in-the-loop escalation for cases where agent disagreement or low confidence signals unreliable output. These architectural requirements are framed as non-negotiable for deploying multi-agent AI in domains where errors carry institutional or societal consequences.
- c. **Researcher Comments** This paper directly validates the design principles underlying our Council's arbitration logic, particularly our use of confidence levels, structured disagreement detection, and a Human Review Requirement flag

triggered when expert consensus falls below threshold. The authors' argument that domain-specific constraints and regulatory requirements must be embedded into multi-agent system design through structured prompting and operational constraints mirrors exactly how our Governance Expert module enforces UNICC's institutional policy requirements within the council workflow.